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# A Method For Valuing High-Risk, Long-Term Investments

## The “Venture Capital Method”

### Introduction

Valuing investments that are characterized by negative early cash flows or earnings, followed by significant projected rewards is very difficult. Yet, investors are confronted frequently with investments whose current value must be estimated in spite of the fact that so much of the reward lies in the future. Examples range from start-up ventures to biotechnology companies to leveraged buyouts.

The objective in this note is to describe a method that has proved useful in valuing such “futures” investments. Variations of the method have been used by venture capitalists (VCs) for many years in deciding how to price venture deals. Indeed, most of the examples in the note will refer to decisions by VCs, though the method is really applicable in many other settings.

The method entails the following steps. First, a company’s net income is projected for some terminal year, say five years from the present. The estimate of net income is typically based on a “success scenario,” that is, one in which the company attains its sales and margin projections. Then a price-to-earnings ratio (PER) is determined that is deemed appropriate for a company that has achieved the measure of success implicit in the forecasted income. Often, this PER is estimated by studying current multiples for companies with similar economic characteristics (e.g., size, profitability, growth rate, capital intensity, risk). The product of the projected net income and the estimated PER is the company’s projected terminal value. This terminal value is then converted to a present value by applying a very high discount rate, typically between 35% and 80% per year. The resulting figure is the estimated current total value as of today. If that total value is \$4 million, to illustrate, and a venture capitalist is being asked to invest \$2 million, then he or she will demand 50% of the value of the company, or one-half of the shares.

As you read this note, you should keep in mind several issues. First, there is no perfect method for valuing high risk, long-term investments. Valuation is part art and part science, and no one can predict the future with certainty. However, decisions must be made, and using analysis to narrow the region of feasible or desirable can be extremely useful. Thus, toward the end of this note, a great deal of attention will be focused on using the method to ask intelligent questions rather than to provide answers. By asking questions that acknowledge our inability to predict the future with certainty, better decisions can be made.

In a broader context, the issues addressed in this note are really related to how markets work. “How are prices determined in the capital markets?” is the key underlying question. The objective in

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Professor William A. Sahlman and Associates Fellow Daniel R. Scherlis prepared this note as the basis for class discussion.

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the note is to discuss some of the key factors that seem to underlie the process by which the capital markets assign values to a given class of investments. The reader's goal should be to come to understand the intuition of that process.

Finally, and most importantly, the issue of valuation must be kept in perspective. Seasoned investors will focus first on the quality of the management team they are backing and on the nature of the opportunity under consideration. *Only after the fundamental decision to invest has been made will the investor turn to the issue of valuation.*

In the first section of this note, the valuation method is applied to a company that is not expected to issue additional shares to attract management or require additional financing. The basic elements of the method are then examined in detail, with particular emphasis on understanding the high discount rates used by VCs, and the assumptions implicit in the use of those rates. The method is then applied to the more complex but realistic example of a company that is expected to issue additional shares over its life, thus diluting the ownership of the original investors.

## The Basic Method

### *The Basic Venture Capital Formula (No Dilution)*

Consider the situation of a venture capitalist who is contemplating a \$3.5 million investment in a company that expects to require no further capital through year 5. The company, MicroElectronics Enterprises (ME2), is expected to earn \$2.5 million in year 5, and should be comparable to companies commanding price/earnings ratios (PER) of about 15. The venture capitalist expects to harvest his investment at that point through sale of his stock to an acquiring company. Assume further that the VC requires a 50% projected rate of return on a project of this risk (venture capital discount rates are discussed in detail below). What price should he pay now for the stock?

| Fact Summary:      |               |
|--------------------|---------------|
| Required IRR:      | 50%           |
| Investment:        | 3.5 million   |
| Term:              | 5 Years       |
| Year 5 Net Income: | \$2.5 million |
| Year 5 PER:        | 15            |

The venture capitalist must own enough of the company in year 5 to realize a 50% annual return on the investment. Thus, at that time his shares must be worth:

$$\begin{aligned}
 \text{Required Future Value (Investment)} &= (1 + \text{IRR})^{\text{years}} \times (\text{Investment}) \\
 &= (1 + .50)^5 \times (\$3.5 \text{ mm}) \\
 &= \$26.6 \text{ million}
 \end{aligned}$$

Now, at that point the company as a whole will be worth:

$$\begin{aligned}
 \text{Total Terminal Value} &= \text{PER} \times \text{Terminal Net Income} \\
 &= 15 \times \$2.5 \text{ mm} \\
 &= \$37.5 \text{ million}
 \end{aligned}$$

For the VC to receive the required \$26.6 million in year 5 out of the \$37.5 million terminal value, he will have to own a corresponding portion of the company's stock. The required percent ownership at that time must be \$26.6/\$37.5, or 70.9%. Since there are not expected to be stock issues between now and year 5, if the VC buys 70.9% now, he will own 70.9% then. When we consider examples with expected dilution, we will see that a higher percent ownership must be acquired in an early financing round to yield a certain terminal ownership.

The above steps can be collapsed into one, the *basic venture capital formula*:

$$\begin{aligned}
 \text{Final Ownership Required} &= \frac{\text{Required Future Value (Investment)}}{\text{Total Terminal Value}} \\
 &= \frac{(1 + \text{IRR})^{\text{years}} \times \text{Investment}}{\text{PER} \times (\text{Terminal Net Income})}
 \end{aligned}$$

Alternatively, you may wish to think about the above formula as:

$$\text{Final Ownership Required} = \frac{\text{Investment}}{\text{Terminal Value} / (1 + \text{IRR})^{\text{years}}}$$

### Computing Shares to Issue and Share Price

When a venture capitalist invests in a company, additional shares are issued, diluting the ownership of the previous investors (who are usually, in a start-up or first-round financing, the venture's founders). Note that "percent ownership" refers to the portion of the total stock that is owned post-money, or after the new shares are issued:

$$\% \text{ ownership acquired} = \frac{\text{new shares}}{\text{old shares} + \text{new shares}}$$

The formula above may be solved to determine new shares, the number of shares to be issued: (The complete derivation is presented in **Appendix 1**.)

$$\text{New shares} = \frac{\% \text{ ownership}}{1 - (\% \text{ ownership})} \times \text{old shares}$$

equivalently, if: total shares = new shares + old shares

$$\text{then} \quad \text{Total Shares} = \frac{1}{1 - (\% \text{ ownership})} \times \text{old shares}$$

Thus, if the VC must own 70.9% of ME2, and there are 1,000,000 shares outstanding pre-money, then an additional 2,433,476 shares must be issued.

The share price is, by definition, the price paid divided by the number of shares purchased:

$$\text{share price} = \frac{\text{investment}}{\text{new shares}}$$

Thus, if \$3.5 million is invested for 2,433,476 shares, the share price is \$1.44.

### Computing Implied Valuation

No matter what method is used for establishing the percent ownership that the VC should obtain for a given investment (that is, for establishing the share price), it is possible to infer a value for the company as a whole. If, for example, one third of a venture is purchased for \$4 million, then the venture as a whole is implicitly assigned a value of \$12 million. In general:

$$\text{post-money valuation} = \frac{\text{investment}}{\% \text{ ownership acquired}}$$

This quantity is referred to as the *post-money valuation* because it is the valuation applicable after the financing. For the ME2 example, above, the post-money valuation is \$3.5mm / 70.9% = \$4.9 million. Not surprisingly, given the method used to derive the 70.9% figure, this is equal to the present value of the venture's total terminal value.

The post-money valuation is the most frequently used "valuation" figure. Note that this \$4.9 million valuation figure includes \$3.5 million of value that has just entered the company in the form of cash.

It is sometimes important to consider the value attributable to only that portion of the company *not* purchased in this financing round. That figure is the *pre-money valuation*:

$$\text{pre-money valuation} = (\text{post-money valuation}) - (\text{new investment})$$

For the ME2 example above, the pre-money valuation is \$4.9mm - \$3.5mm = \$1.4 million. The pre-money valuation is that portion of the post-money valuation attributable to stock held by founders, employees, and previous investors.

The *carried interest* of a given shareholder (or group) is the valuation of that portion of the company owned by that party, after the financing. Thus, the post-money valuation may be broken down into the carried interests of the various parties. For any given shareholder:

$$\text{carried interest} = (\text{post-money valuation}) \times (\% \text{ ownership post-money})$$

The carried interest of the new investors is equal to the value of their investment. In the case of a start-up, the founders' carried interest constitutes the entire pre-money valuation; it is the value assigned to their technology, and to their hard work, thus explaining its other name: *sweat equity*. Using the above formula, the founders' carried interest is \$4.9mm x (1 - 70.9%) = \$1.4 million, which is of course equal to the pre-money valuation.<sup>1</sup>

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<sup>1</sup>Another definition of *carried interest* subtracts a shareholder's previous investments from his portion of the post-money valuation. For example, if the founders had contributed \$100,000 for their initial equity, then their carried interest would be

The above items can also be expressed in terms of shares:

$$\text{post-money valuation} = (\text{new price}) \times (\text{total shares})$$

$$\text{pre-money valuation} = (\text{new price}) \times (\text{old shares})$$

$$\text{carried interest} = (\text{new price}) \times (\text{old shares})$$

### *Working with Implied Valuations*

In general, these calculations, and calculations of ownership percentages, include options and warrants issued. Typically, warrants are issued and held in reserve as incentives for management. Since managers receive these warrants for free or for a price well below their fair value, they are treated as stock for the purpose of calculating valuations.<sup>2</sup> Valuations that include options and warrants as if they had been already exercised are said to be *fully diluted*.

Although the valuation assigned in a financing can be clearly stated in algebraic terms, its degree of ultimate reality should be treated with some skepticism. A VC who buys stock in a venture is not simply exchanging cash for stock; he also commits expertise, connections, and direction in exchange for the stock and for the option to invest further or to abandon the project as it unfolds. A one-time passive investor would probably set a very different price than the long-term venture capitalist.

In addition, the shares purchased by the investors are seldom strictly comparable to stock owned by founders. Typically, the founders hold common stock, while the investors own preferred stock. Preferred stock can carry various rights, the most frequent being liquidation preference, which grants the investor a return of all (or sometimes a multiple of) his investment before any proceeds from sale or liquidation become available to the common stockholders. Owners of preferred stock can also receive dividends, either paid on a current basis or accrued.

There is another crucial difference between investors' and founders' stock. A purchaser who paid the full "valuation" for all the founders' stock and obtained 100% ownership would be left with a company whose management had been stripped of incentives to create long-term value. The founders' stock ownership creates value; it is unlikely that any purchaser would allow them to sell their stock and walk away from the venture. For this reason, VCs generally prevent management from selling their stock, by means of gradual vesting (transfer of ownership) of stock ownership or options, or by negotiating automatic buy-back agreements that give the VCs the right to buy the stock of a manager who sells, but at prices that are discouragingly low in earlier years.<sup>3</sup>

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(\$1.4 mm - \$0.1 mm) = \$1.3 million. This represents the amount by which the valuation of a shareholder's portion increases due to an increase in company valuation. Note that, under this definition, if an investor had bought in at a price higher than the current round's price, then his carried interest would be negative.

<sup>2</sup>An *option* is the right to buy a certain number of shares of existing stock at a certain exercise price. A *warrant* is similar, except the new stock is issued when the warrant is exercised. The fair value of an option or warrant can be calculated based on its exercise price, expiration date, the level of interest rates in the economy, and the price and expected price volatility of the underlying stock. Warrants are generally worth less than comparable options because of the dilution associated with the issuance of new shares. The value of a very long-lived warrant with a very low exercise price approaches the value of the stock itself. Since management warrants generally match those characteristics, it is reasonable to approximate their value by treating them like a reserve of actual stock.

<sup>3</sup>A *lock-up* is a similar provision, except that it is the company, not the VC, that has the right to buy the manager's stock at a pre-arranged price.

## Discussion of the Elements of the Venture Capital Method

In the sections below, we explore further the key elements of the venture capital method: the discount rate, the terminal value, and the investment size. Because the determination of each of these elements (especially the discount rate) varies according to the stage of development of the venture, in the first section we discuss the characteristics of the stages that VCs generally recognize.

### *Stages of Investment*

Venture capitalists distinguish between certain generic stages of a venture's life cycle. Bear in mind that the definitions below are approximate; VCs vary widely in their use of these terms. The discount rate ranges represent a rough consensus of venture capitalists, and will be discussed in more detail in the following section.

**Seed Financing** is the earliest stage of funding. A small investment (typically \$25,000 to \$300,000) is made to support an entrepreneur's exploration of an idea. Often there is no business plan, an incomplete management team, and little assurance that the basic technology is feasible. Sometimes, when the product technology is well established, seed money is raised simply to finance the recruitment of key management and the writing of a business plan, both of which are generally necessary for startup funding. Seed investors expect to provide basic business advice, and perhaps even office facilities, for their entrepreneurs. Seed investors often apply discount rates of over 80% to the projects in which they invest.

**Startup Financing** entails the commitment of more significant funds to an organization that is prepared to commence operations. A start-up should be able to demonstrate a competitive advantage. Most high-technology firms should have a product in prototype form embodying a proprietary technology. A research-oriented venture, such as a biotechnology firm, might instead exhibit an impressive research staff. Low-technology ventures, such as specialty retailing or entertainment, should have a powerful concept with pre-emption advantages and a superior management. Investors in startup ventures frequently provide assistance to management in recruiting key personnel, establishing sound management practices, and providing access to suppliers, banks, and potential customers. Start-up investors apply discount rates of 50% to 70%.

**First-Stage Financing** is provided to on-going businesses. A first-stage company is generally not profitable, but it normally has an established organization, a working product, and, preferably, some revenues. First-stage funds are usually used to establish a company's first major marketing efforts, and to hire sales and support personnel in anticipation of higher sales volume. Often, funds are also applied to product enhancements or product line expansion. First-stage investors attempt to monitor closely a venture's head count, ensuring that staffing levels correspond to attainable sales levels. They often become more actively involved as problems develop in production or sales, and are prepared to replace key managers as necessary, sometimes filling in key positions themselves while searching for new managers. Discount rates applied to a first-stage venture are generally 40% to 60%.

**Second-Stage Financing** is typically provided for working capital and fixed asset needs to support the growth of a company with active production, sustainable sales, and, preferably, some profits. Whereas earlier-stage funds were largely dedicated toward proving a venture's viability, second- and later-stage capital is oriented towards the expansion of a tested contender. Since the capital invested in the later stages is more likely to pay for assets rather than operating expenses, it is more readily recoverable in the event of liquidation, thus lowering the overall risk to the investors. Second-stage investors do not generally expect to become actively involved in problem-solving as

often as first-stage investors. They do monitor performance closely, generally by comparison to a business plan. Discount rates for second-stage investments range from 30% to 50%.

**Bridge Financing** is intended to carry a company until its initial public offering (IPO). Although an IPO is not yet appropriate due to market timing or the size and performance of the company, it is generally expected within a year after the bridge. Bridge investors might provide funds to satisfy ongoing capital needs, with the expectation of selling out again in the IPO as part of a secondary offering (an offering of shareholders', as distinct from company, stock). Alternately, bridge investors might apply some or all of their funds to buy out early-stage investors who are anxious to liquidate their holdings. Such an investor often expects to hold the stock past the IPO date, as a long-term investment. Bridge investors are generally passive investors. They apply discount rates of 20% to 35%.

**Restart Financing**, also known as emergency or sustaining financing, is raised for a troubled firm, at a price significantly below that of the previous round. Although the venture is performing well below expectations, the round will be priced low enough to offer a high expected rate of return, one result of which is likely to be substantial dilution of any previous investors that do not participate in the restart financing.

### *The Venture Capitalist's Discount Rate*

The rate used by the venture capitalist to discount expected future value to the present has an enormous effect on implicit valuation, and thus on the VC's required ownership and the share price. The 50% rate used in the ME2 example above is typical of first-round venture capital financings. The range of discount rates applied at different stages (as per the preceding section) is from 20% to 70%. To gauge the reasonableness of these discount rates, consider the average annual nominal returns realized on a variety of investments from 1986 to 2002:

| Assets                     | Nominal Annual Return |
|----------------------------|-----------------------|
| Venture Capital            | 13.9%                 |
| Large Company Stocks       | 12.5                  |
| Small Company Stocks       | 12.3                  |
| Long-Term Government Bonds | 10.2                  |
| Real Estate                | 8.0                   |
| Foreign Common Stocks      | 4.4                   |
| Gold                       | -0.2                  |

Source: Ibbotson Associates, VentureXpert, Datastream, COMEX.

In light of these historical returns, the venture capitalist's insistence on such high rates of return rate seems difficult to justify. Furthermore, since the venture capitalist anticipates receiving essentially his entire return from the terminal value, a high discount rate has a more extreme effect than it would on a company that provided interim returns.

To illustrate the sensitivity of present value to the use of high discount rates, consider two bonds. Bond 1 is a five-year interest-bearing bond paying \$10 interest per year, and returning its \$100 principal in year five. Bond 2 is a zero-coupon bond, distributing only a \$100 payment in year five.

The zero-coupon bond, of course, has a lower present value at any discount rate greater than zero, since it pays no interest. However, its present value drops more abruptly with an increase in discount rate, as indicated by the “relative present value” ratio indicated below. The second bond is more comparable to a venture capital investment in the sense that it does not return value until the terminal year. The first bond, stable in value, is comparable to a mature public company that periodically returns cash to its investors.

| Discount<br>Rate                                    | Year |    |    |    |     | Present<br>Value | Relative<br>Present<br>Value |
|-----------------------------------------------------|------|----|----|----|-----|------------------|------------------------------|
|                                                     | 1    | 2  | 3  | 4  | 5   |                  |                              |
| Bond 1) Five Year, Interest-Bearing with 10% Coupon |      |    |    |    |     |                  |                              |
| 15.0%                                               | 10   | 10 | 10 | 10 | 110 | (83.24)          | 100.0%                       |
| 25.0                                                | 10   | 10 | 10 | 10 | 110 | (59.66)          | 71.7                         |
| 35.0                                                | 10   | 10 | 10 | 10 | 110 | (44.50)          | 53.5                         |
| Bond 2) Five Year, Zero-Coupon, Balloon Payment     |      |    |    |    |     |                  |                              |
| 15.0%                                               | 0    | 0  | 0  | 0  | 100 | (49.74)          | 100.0%                       |
| 25.0                                                | 0    | 0  | 0  | 0  | 100 | (32.77)          | 65.9                         |
| 35.0                                                | 0    | 0  | 0  | 0  | 100 | (22.30)          | 44.9                         |

The enormous gap between historical realized returns on investments and the fact that valuations are so sensitive to discount rates suggests a question: why do VCs use such high discount rates? In order to address this issue, five possible determinants of discount rates are examined below: the base return, systematic risk premium, a liquidity premium, compensation for value added, and a cash flow adjustment. Also discussed is how each of these factors change over time.

**Base Rate of Return** For a risk-free investment, one with no chance of default, the appropriate discount rate would be the government bond rate. Historically, this “base rate” compensates for inflation and provides a small additional “real” return as a reimbursement for the use of funds. For the period from 1926 to 2002, the average real return on short-term riskless assets was 0.8% and 2.7% on long-term government bonds (see Exhibit 1). Obviously, venture capital investments are perceived as far more risky than government bonds. The issues are how much riskier, and how much to charge for the extra risk.

**Systematic Risk Premium** According to the capital asset pricing model (CAPM),<sup>4</sup> a stock’s rate of return reflects the market’s perception of its systematic risk, defined as the sensitivity of the return on the stock to the return on the market as a whole. This risk cannot be diversified away and investors demand a premium return to bear it.

With respect to investments such as those made by venture capitalists, total risk is very high. That total risk (defined as uncertainty about future return) can be decomposed into two parts: systematic (or market) and unsystematic (or non-market) risk.

Certainly, venture capital investments contain a high amount of unsystematic risk. The uncertainty as to whether a technology will work or can be commercialized, is probably not market-related. However, since unsystematic risk can be diversified away, investors can expect no extra return for bearing unsystematic risk. The fact that venture capitalists typically hold many different securities in their portfolio suggests they understand the necessity of diversification. Moreover, the

<sup>4</sup> For background reading on the capital asset pricing model, see Behzad Razavi, Stewart C. Myers and Richard Brealey, *Principles of Corporate Finance* (Seventh Edition), New York, NY, 2002.

indirect owners of most VC funds—large institutional investors—also take full advantage of the risk reduction benefits of diversification.

Thus, since unsystematic risk is effectively diversified away and commands no extra return, the issue remains as to the degree of systematic risk present in venture capital investments.

An argument can be made that the value of these investments is highly vulnerable to market conditions generally, especially those in effect when the VCs attempt to liquidate their investments via public offerings or in acquisitions by larger companies. Terminal value estimates are highly sensitive to PERs, and the PER a given company can command will be highly dependent on PERs in the market at large. Therefore such investments are likely to have a relatively high level of systematic risk.

To illustrate first the sensitivity of the VC's returns to terminal value changes, consider again the two bonds from above. Assume that the bonds are purchased in year 0 for their present value, as determined in the table above. Given a 50% reduction in the final-year value (which might be the result of a 50% reduction in the price/earnings ratio applied to a given net income) the second bond, whose only return is in the final year, suffers a much greater reduction in the realized rate of return:

| Original<br>Disc Rate                               | Year    |    |    |    |    |         | NPV at<br>Original<br>Disc Rate | Actual<br>Rate of<br>Return |
|-----------------------------------------------------|---------|----|----|----|----|---------|---------------------------------|-----------------------------|
|                                                     | 0       | 1  | 2  | 3  | 4  | 5 (adj) |                                 |                             |
| Bond 1: Five-Year, Interest-Bearing with 10% Coupon |         |    |    |    |    |         |                                 |                             |
| 15.0%                                               | (83.24) | 10 | 10 | 10 | 10 | 55      | (27.34)                         | 3.4%                        |
| 25.0                                                | (59.66) | 10 | 10 | 10 | 10 | 55      | (18.02)                         | 13.0                        |
| 35.0                                                | (44.50) | 10 | 10 | 10 | 10 | 55      | (12.27)                         | 22.6                        |
| Bond 2: Five-Year, Zero-Coupon, Balloon Payment     |         |    |    |    |    |         |                                 |                             |
| 15.0%                                               | (49.72) | 0  | 0  | 0  | 0  | 50      | (24.86)                         | 0.1%                        |
| 25.0                                                | (32.77) | 0  | 0  | 0  | 0  | 50      | (16.38)                         | 8.8                         |
| 35.0                                                | (22.30) | 0  | 0  | 0  | 0  | 50      | (11.15)                         | 17.5                        |

Even before an IPO or acquisition, the fortunes of a VC-backed company are likely to be highly sensitive to market conditions. High-technology companies are often providers of productivity tools to more-established industries, which often reduce capital expenditures during hard times. Furthermore, when any industry goes through a contraction, the smaller players, who are often highly specialized and thinly capitalized, tend to be the first to fail. Start-up companies often have very high fixed costs relative to total costs. Thus, they will be more likely to suffer losses and cash shortages given a proportional reduction in revenues.

Also, economic contractions are likely to occur at the same time as depressed stock markets. Generally low stock prices in turn tend to occur at the same time as weak IPO markets (see **Exhibits 2, 3, and 4** for evidence). As a result, ventures faced with tough business conditions might be unable to raise capital in the public market on acceptable terms. Loss of access to the public stock market might exacerbate the demand for money on the venture capital fund since all portfolio firm financial needs must be met from internal sources or from further funds from the VC portfolio. Such funds might not be forthcoming.

The volatility of values assigned to young companies is also immediately evident upon examining the valuation and performance of DataStream Information Technology Index of small public high-growth companies, relative to the Standard & Poor's 500 (see **Exhibit 2**). When these data are used to calculate a beta for the DataStream Information Technology Index, the resulting estimate is 1.36, a

relatively high figure. This estimate suggests that the Index is 36% more volatile than the market in general.

A similar regression of the excess return on the Venture Economics Post Venture Capital Index (an index of publicly traded companies that had been backed by venture capital) on the excess return on the S&P 500 reveals a beta estimate of 1.65.

Finally, studying the behavior of other stock prices reveals a similar pattern. For example, the return on the Russell 2000 Growth Index from 1985 to 2002 is 33% more volatile than the S&P500, as measured by the ratio of standard deviations of monthly return. These results are portrayed in **Exhibit 2**.

This analysis suggests that the systematic (and unsystematic) risk of venture capital-type investments can be high. The next issue is: how much of the venture capitalist's high discount rate can be explained by systematic risk? A stock's systematic risk can be gauged by its *beta*, a measure of covariance with market movements. According to the capital asset pricing model (CAPM), a stock's expected rate of return should equal:

$$(\text{risk-free rate}) + (\text{beta} \times \text{market-risk premium})$$

Given an estimated risk-free rate of 4% and an estimated market risk premium of 6%,<sup>5</sup> the beta of a venture capital investment would have to be 7.67 in order to justify using a discount rate of 50% [= 4% + (7.67 × 6%)]. Yet, among S&P 500 stocks, only 1 had a beta over 4.0 and only 10 had a beta between 3.0 and 4.0 (see **Exhibits 5, 6, and 7**). Even if the correct beta estimate were 4.0, then the required rate of return would be 28%, not 50%.

The only logical conclusion one can draw from these data is that, while systematic risk is high, systematic risk alone cannot be used to justify very high discount rates. What then explains the rates applied by venture capitalists? Part of the answer can possibly be found by exploring the consequences of illiquidity for such investments.

**Liquidity Premium** Stock held in private companies differs from publicly traded stock in a number of dimensions. Most important, such investments are characterized by illiquidity, defined as the inability to convert the holdings to cash at their full value in a reasonable period of time. A publicly traded stock can be sold at the market and cash received almost immediately. Stock in a privately held company cannot be sold so easily for a number of reasons: first, the amount of credible information available (e.g., audited statements) is far less; second, there are legal restrictions on the sale of unregistered securities; and, finally, there are a relatively small number of potential buyers. These factors suggest that, holding all other things constant, a buyer should pay less for the private stock.

This prediction is borne out in the markets. Typically unregistered stock that is sold commands a price well below that for registered shares. Discounts up to 50% are applied. Presumably, a major determinant of the magnitude of the discount is the fact that the investment is less liquid.

Thus, the high discount rate applied by venture capitalists can in part be justified, given the illiquid nature of the investment. However, this argument can only be taken so far. There are many potential investors for whom illiquidity, per se, is not a major issue. A good example is a pension fund, which can afford to invest in projects that have long payback periods and are illiquid before

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<sup>5</sup> The use of 6% should not be interpreted as "truth." The average exchange return on the market for the past 51 years was 8.2% (see **Exhibit 1**). However, some evidence suggests for lower risk premia, perhaps under 4%. To the extent the latter figure is correct, the beta would have to be higher to justify high rates.

harvest. If a very large discount were applied to illiquid investments, then investors like pension funds would increase the allocation of assets to such opportunities, driving up the prices paid and lowering the discount. This process of market equilibration will likely result in justifiable liquidity premiums that are not nearly large enough to justify 50-80% discount rates.

Given the gap between the level of justifiable expected return based on compensation for risk and compensation for illiquidity, the rationale behind high venture capital discount rates must be sought elsewhere. We turn next to the issue of compensation for “value-added” by the venture capitalist.

**Value Added** Arguments about discount rates involving systematic risk and liquidity premiums fit well within the perspective of capital market theory. According to that theory, capital is a commodity and investments and investors are undifferentiated. Venture capitalists, however, are active investors, and generally do not passively hand over capital for management to do with as it pleases. Their efforts in hiring, firing, and advising managers are directed toward increasing the value of their stock holdings, and thus are necessarily directed toward increasing the total value of the portfolio company. Thus, the high rate of return demanded by venture capitalists could be viewed partly as a compensation for services rendered. Presumably, this premium for added value is reflected in the fund manager’s share of the fund’s profits, rather than being passed on to the investors, since it is the fund manager who adds the value.<sup>6</sup>

There are several examples of venture capitalists making outstanding contributions to ventures through their actions or simply through their presence. For example, venture capitalist Ben Rosen demanded, and realized, a generous rate of return on his investment in the start-up Lotus Development Corporation. However, it is not clear that Lotus would have succeeded without Rosen’s expertise and his credibility in the microcomputer industry.

Given the salutary effect of Rosen’s reputation and efforts on Lotus’ cash flows, the true cost of Rosen’s capital to Lotus was significantly lower than the return Rosen specified or received.

Thus, companies receive more than money when they accept funds from VCs. The value added by the non-capital contributions of the investor are compensated. In the case of the typical venture capital firm, the compensation takes the form of a 20% override on the portfolio given to the general partners. Holding all other things constant, the return required on any given investment must be grossed up to reflect the compensation to the venture capital managers. Thus, if the required return on the investment were 20%, then the venture capital fund with a 20% override would have to demand 25% in order to have the limited partners attain their required 20% net return.<sup>7</sup>

The extra return required or expected to compensate for value added need not necessarily bear a direct relationship to the size of the equity override. This is so because in certain cases, the value added may be very large and in other cases small, regardless of the overall size of the equity override.

**Cash Flow Adjustment** The three factors above help explain why a venture capital portfolio as a whole should provide a return higher than the market rate. The investor and, therefore, the general partner, is exposed to relatively high systematic risk, and has little opportunity to liquidate his holdings for an extended period of time. In addition, the VC is a value-added supplier of capital.

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<sup>6</sup>Typically, private venture capital funds are structured as limited partnerships. The investors are limited partners (LPs); they are passive participants in the funds, and their liability is limited to the amount of their investment. The managers of the fund are general partners (GPs). They are usually entitled to a percentage of the fund’s profits, although occasionally the LPs must receive a certain minimum rate of return before the GPs participate. Although a partnership’s GPs are exposed to unlimited liability, the managers of a VC fund generally attempt to manage their financial affairs so as to protect accumulated wealth.

<sup>7</sup>The required gross-up would likely be higher to reflect the 2-3% VC management fee.

However, there still remains a wide gap between the average long-term venture capital portfolio return of 25%, and the usual discount rate applied to a first-round investment, generally 50%.

Perhaps the 50% discount rate represents the VC's perception of a fair compensation for systematic risk, lack of liquidity, and value added. If this were so, the 25% actual portfolio return would represent a disappointment to the VC, the result of an unexpectedly low success rate among the portfolio companies. To the contrary, VCs have generally been satisfied with average yields of 25%, as has been reflected in the stability of that rate, the lack of exit from, and the high rate of entry into the venture capital industry over the last 10 years.<sup>8</sup> A fund with a return as high as 30% is considered quite successful, and its investors prove eager to reinvest, hoping for a repeat of that performance.

Recall, from the basic venture capital valuation formula, that the ownership that a VC claims is based on the future value of his investment (equal to the investment amount accumulated at the appropriate discount rate) divided by the forecasted terminal value of the company. If the VC, who *expects* a 27% return, applies a 50% rate in order to derive the appropriate ownership, he is implicitly asserting that the forecasted terminal value is not the terminal value he really expects. Since the VC knows what his own investment is, if he is applying a higher than expected discount rate, he must be expecting the actual terminal value to be lower than forecasted.<sup>9</sup>

If the venture capitalist were to use the expected (rather than the forecasted) terminal value, he could apply the expected portfolio rate of return. If the VC expects the portfolio return to be 30% over five years, applying a 50% discount rate is equivalent to adjusting the forecasted terminal value (TV) by halving it:

$$\frac{\text{Forecasted TV and High Discount Rate}}{1.50^5} \times \text{TV} = \frac{\text{Expected TV and Expected Rate of Return}}{1.30^5} \times (\text{TV}/2.0)$$

Thus, if 30% is the expected return, the forecasted TV is adjusted by dividing by a factor of 2.0. The VC, who uses the forecasted cash flow in his calculations, adjusts it by using the high discount rate. The table below shows the adjustment factor implicit in using various discount rates, given a range of expected rates of return:

|               |     | Expected Portfolio Return |     |                   |     |     |     |
|---------------|-----|---------------------------|-----|-------------------|-----|-----|-----|
| Discount Rate |     | 15%                       | 20% | 25%               | 30% | 35% | 40% |
|               | 20% | 1.2                       | 1.0 | 0.8               | 0.7 | 0.6 | 0.5 |
|               | 30% | 1.8                       | 1.5 | 1.2               | 1.0 | 0.8 | 0.7 |
|               | 40% | 2.7                       | 2.2 | 1.8               | 1.4 | 1.2 | 1.0 |
|               | 50% | 3.8                       | 3.1 | 2.5               | 2.0 | 1.7 | 1.4 |
|               | 60% | 5.2                       | 4.2 | 3.4 <sup>10</sup> | 2.8 | 2.3 | 1.9 |
|               | 70% | 7.1                       | 5.7 | 4.7               | 3.8 | 3.2 | 2.6 |

<sup>8</sup> See Note on the Venture Capital Industry—Update (1985) (HBS Case #9-286-060) for confirmation of this fact.

<sup>9</sup> The issues explored in this section were first discussed in Robert Keeley, "Risk (Over)Adjusted Discount Rates: The Venture Capitalist's Method," unpublished (1985). See also Howard H. Stevenson, Daniel F. Muzyka and Jeffry A. Timmons, "Venture Capital in Transition: A Monte-Carlo Simulation of Changes in Investment Patterns," *Journal of Business Venturing* (Spring 1987), pp. 103-122.

<sup>10</sup> For example: if the expected portfolio return is 25%, applying a 60% discount rate is equivalent to dividing the expected cash flow by 3.4.

The VC knows, from experience, that only about 25% of his investments will meet or exceed their projections, that another 25% will fail to return all of the invested capital, and that the remaining 50% will produce modest returns on investment, perhaps only managing to cover interest and principal on subordinated debentures (or dividends and redemption on preferred stock), without reaching a level of success adequate for a major gain from the sale of the company. In order to produce acceptable returns on a portfolio of such companies, those that do achieve their projections must return a substantial profit to the VC. The 50% discount rate applied to a first-round company such as ME2 will produce the 25% average portfolio return that the VC expects, given the distribution of individual returns cited above:

| Performance Category | % of Total Investments | Cash Multiple Over 5 Years | Rate of Return    |
|----------------------|------------------------|----------------------------|-------------------|
| Successful:          | 25%                    |                            |                   |
| Exceed plan          | 10%                    | 13.00                      | 67% <sup>11</sup> |
| Meet plan            | 15                     | 7.60                       | 50                |
| Sideways             | 50%                    | 1.75                       | 12                |
| Loss                 | <u>25</u>              | <u>.25</u>                 | <u>(24)</u>       |
| Total                | 100%                   | 3.38                       | 28%               |

The above algebra notwithstanding, VCs probably do not apply the discount rates they use based on careful historical analysis of distributions of returns. VCs apply high discount rates because they work. The successful venture capitalists are those who have demanded rates high enough to compensate for a venture's likely performance shortfall relative to forecast, but not so high as to force the managers of too many potential investments to seek alternate funding. By experience, the venture capital community has learned that applying high discount rates to forecasted performance can compensate for the fact that earnings forecasts often are not realized, and thus that system remains popular.

The VC who uses a 50% discount rate for a group of investments is assuming that each investment in that group has a similar probability of achieving the success, sideways, or loss scenario, and also that each investment has a similar relative payout in each of those scenarios. In fact, an investment in a capital-intensive manufacturing company is likely to pay out more under liquidation (the "loss" scenario) than an investment in a salary-intensive research company. In order to treat explicitly these varying payouts under each scenario—and to assign different probabilities of success or failure to different ventures—many venture capitalists apply the First Chicago Method, which is presented in **Appendix 3**. That method applies a lower, expected, discount rate to a weighted average cash flow, which is derived from forecasts of each scenario.

### *Changes in Discount Rates Over Time*

The discussion of the elements of the venture capital method began with the identification of certain stages that VCs identify, ranging from seed financing to bridge financing. For each stage, the different discount rate is typically applied. The next section contained an analysis of the reasons why venture capitalists might apply high discount rates. These were: (1) to provide a base return at least equal to the risk free rate; (2) to provide compensation for risk bearing; (3) to compensate for

<sup>11</sup>For example: 10% of venture capital investments return 13 times the original investment after five years, for a 67% annual rate of return.

illiquidity; (4) to take into account the value added by the investor; and (5) to adjust future cash flow projections to reflect experience and skepticism.

One unresolved issue is how justifiable discount rates change according to the stage of development of the company. An argument can be made that each new stage that a company attains warrants a decline in the discount rate premium over the risk-free rate.<sup>12</sup>

For example, it seems likely that the systematic risk of a company will decline as it grows in size, as its growth rate declines, and as the relative level of fixed costs declines. So, too, will total risk likely decline as the company expands, though the effect on discount rates is not clear.

The same basic argument applies to the premium for illiquidity and for value added. The larger and more successful the company, the more potential buyers there are. The existence of more buyers suggests greater potential liquidity, as defined earlier.

As companies evolve and grow, they are usually less reliant on value added by investors. They have an experienced management team in place. They need not rely so much on the credibility of the funds provider. Generally, the VCs are less active as companies prosper.

With respect to the “fudge factor” phenomenon, a strong case can be made that the justifiable discount applied to future forecasts declines with the passage of time. Four years into a venture, the ability to forecast revenues and profits should be far greater than on day 1.

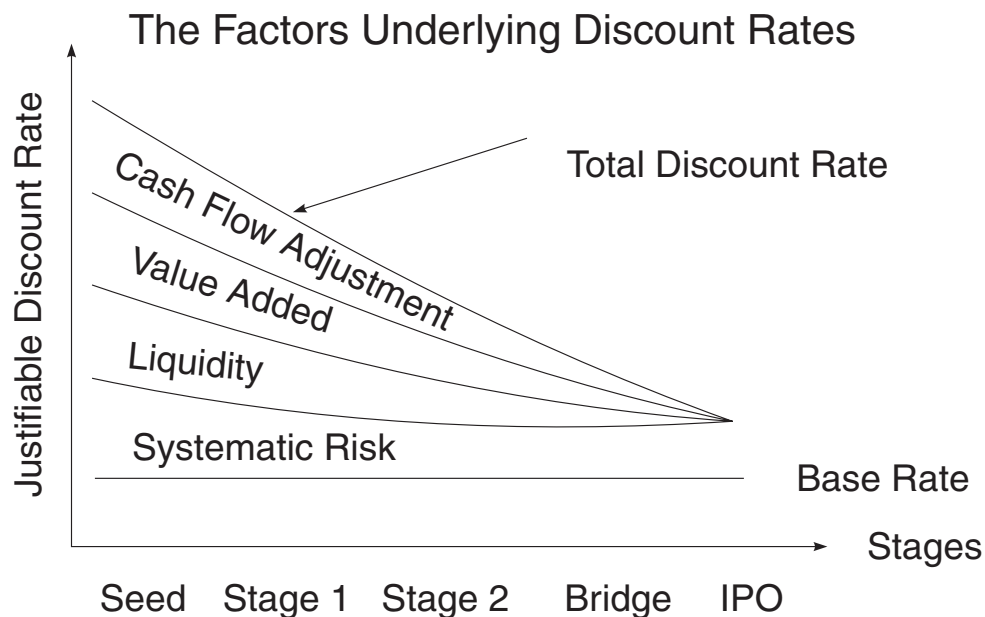
In total, it seems likely that the discount rate appropriate for companies that attain new, higher states will decline asymptotically to the level of return appropriate for a publicly traded company without value added investors. By the time the company goes public, the stock is liquid, has lower systematic risk, has little value added from the investors, and has a lengthy track record on which to base projections.

If, on the path to success, the company stumbles, then the process begins again, and the appropriate discount rate will increase accordingly. This is why restart financings are often associated with high discount rates.

These issues are portrayed graphically below:

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<sup>12</sup> See **Appendix 4** For a further discussion of the implications of applying different discount rates at different stages.



### *The Terminal Value Calculation*

The primary source of profits for the venture capitalist is, ultimately, the cash received upon harvest. Since that terminal value is the fundamental input to the valuation process, it must be believable. It is very easy for the entrepreneur to develop a forecast designed to justify a desired valuation: knowing the required investment, the desired ownership to be sold, and the VC's discount rate, the necessary future earnings and sales can then be calculated and presented as forecasts. These forecasts can then be justified, in terms of market share, by defining the market broadly enough to provide a small, "conservative"-looking percent share. (Some venture capitalists refer to the "China Syndrome": if the potential market is defined to include all the people in China, then great riches can be forecast based on a tiny market share.) Thus the investor must develop his own forecasts as a requisite for valuation.

The usual terminal valuation method used by venture capitalists is the application of a price/earnings ratio (PER) to projected net earnings. This is the measure commonly used in the public stock market, which is where the VC generally hopes to harvest the venture by means of an initial public offering or an acquisition in the terminal year.

If another means of harvest is expected, then the valuation method that best predicts that harvest value should instead be used. For example, retailing businesses are frequently valued using a value-to-revenue ratio. This "revenue multiplier" is also used in situations where no PER is available, as for example in an emerging industry where none of the publicly held competitors have achieved profitability yet or where net income is not necessarily indicative of actual cash flows. Real estate developers work primarily with capitalization rates. Cable TV operators and magazine publishers frequently work with price-per-subscriber or price-to-cash-flow rules of thumb. A VC expecting to sell a venture to such an operator would do well to use the potential purchaser's valuation method in predicting the likely terminal value.

In some industries, there may be no readily available valuation measure. Public internet companies often have no profits to apply a PER to, and revenues, if any, that are too low to yield a reasonable revenue multiplier. Internet analysts generally value these firms by projecting the total

future market for the various anticipated products, and assigning a market share to each firm. The terminal value is estimated by applying PERs and anticipated profit margins to these sales forecasts. However, there is no history of appropriate margins and PERs for these analysts to draw on. (See **Exhibit 8** for current internet valuation measures.) Thus, some analysts examine new measures in search of a generalizable pattern.

By judiciously applying a variety of different techniques, each as a check on the others, the investor faced with a difficult valuation task can increase his confidence in a terminal value estimate. In addition, the plausibility of a terminal value should be checked by comparison to other, external financial factors. For example, the projected revenues of a company that provides computer components should not outstrip the expected revenues of the computer manufacturers that the venture will supply. Those revenues should be compared to overall industry forecasts, and the implied market shares should be reasonable. Terminal profit margins should be plausible; if a venture is expected to generate fabulous returns on revenues and capital, new competitors will naturally arise, reducing those returns.

Even after a terminal value forecast passes these checks, at the level of the individual firm, the expected performance of known competitors should be examined as well. If 20 firms each expect a 10% share in an emerging market, there are bound to be unpleasant surprises. If, to justify current valuations and support a reasonable growth rate, those 20 firms would have to produce aggregate revenues comparable to the United States Gross National Product, it is safe to assume that the industry has suffered drastic over-investment.

The general point about estimating terminal values is simple: the analyst must engage his or her brain and must run a number of checks. Among the critical questions are the following:

1. Do the company's revenue forecasts make sense, given:
  - a) the overall industry projections?
  - b) the level of entry and competition?
  - c) the internal ability of the company to sustain the growth rate?
2. Do the company's margin forecasts make sense, given:
  - a) the level of entry barriers now and expected in the future?
  - b) the relative bargaining power of suppliers and customers?
  - c) the threat of substitutes?
  - d) the intensity and form of current and projected competition?
3. Does the terminal valuation, given agreement on terminal sales and margin forecasts, make sense, given the current level of valuations in the market for:
  - a) liquidation?
  - b) initial public offering?
  - c) acquisition by another company?

To address these questions, the analyst should gather relevant current information, as well as studying past history. In particular, it will be useful to search for historical analogies. Has any company ever grown from \$0 to \$1 billion in revenues in five years while sustaining 18% net margins? Has any OEM supplier ever been able to attain 12% after-tax margins? These are the kinds of questions that can and should be asked. For some information on ranges of valuation and performance data, see **Exhibit 9**.

Calculating a terminal value in year five does not necessarily imply an intention to harvest (e.g., convert to cash) the investment at that time. Such a decision would be based on a number of factors

like market pricing, marginal tax rates, and the company's performance trends. However, it is very useful to estimate what the consequences are of selling shares at some future point in time.

### *The Size of the Investment*

The basic venture capital pricing formula requires an appropriate risk-adjusted discount rate, a terminal value (usually a terminal net income and a price/earnings ratio), and a specified level of investment. Of course, only the investment amount is known with certainty as of the financing.

That figure is determined by a variety of factors. The venture needs a certain amount of capital to reach the next stage. What defines the next stage, and how much it will take to get there are both dependent on the venture. As thresholds are crossed in which prototype viability, scale manufacturing, or market acceptance are demonstrated, major concerns are dissipated and a new round of funding at a higher share price becomes viable. The venture's management must be sure to raise enough capital to ensure that the next stage can be reached, but they will be reluctant to raise excess funds at the low early-stage share price. For example, consider a first-stage biotechnology firm that expects to "burn" \$50,000 a month in research expenses during the two years of product development and testing required to achieve FDA approval. That approval represents a major milestone, and will trigger the next round of funding, which will support product marketing. Thus, this company will attempt to raise over \$1.2 million.

To estimate the future financial needs of a company, a pro forma forecast can be prepared. The pro forma will indicate the total annual external financial need of the company. Then two decisions must be made. The first is determining to what degree the venture can raise capital from non-equity sources (e.g., a revolving line of credit based on receivables and inventories). The second decision is identifying the stages at which money will be raised in the future. The identification of stages will depend on the attainment of underlying key developmental goals.

Having determined total future external financing needs, the likely sources, and the appropriate timing of financing, the analyst can then assess the reasonableness of these estimates. Are the numbers feasible, given the strengths and weaknesses of the company and its current and future competitors? Are the financial ratios sensible given ratios for other companies in the industry or in similar industries? These are the kinds of questions that can and should be posed.

While the total amount of funding to be raised depends on the company's needs, how much of that amount a given venture capitalist will contribute depends on that VC's needs. For example, a small fund may wish to diversify its investments across several firms in each of several emerging industries. That fund may, by policy, limit any given investment to \$500,000 (in which case two or more funds would form a syndicate to pool the required \$1.2 million for the above biotech firm), whereas a larger fund wishing to concentrate in biotechnology might be happy to make the full investment. Also, VCs often plan on participating in a venture's future rounds. Thus, a VC willing to place up to \$500,000 with a venture will often invest only \$250,000 in the first round in order to conserve \$250,000 for a future round.

Although VCs generally have a *maximum* investment level, as determined by diversification needs, they will also have a *minimum* level for any investment, determined either by the size of the investment (e.g., no less than \$500,000 placed in any given investment) or by the expected return (e.g., the expected harvest must exceed \$5 million, regardless of investment size). In order to realize value from their investments, the fund's managers need to commit time to board meetings, consultation with management, and other monitoring activities. Because of the number of competing opportunities for the VC to add value, there is a substantial opportunity cost (or shadow price) to the

VC's time. Thus, an investment must be large enough to justify the commitment of time it entails; if the number of investments per manager becomes excessive, it will become impossible to add value to any of them. In addition, if a venture capitalist will be one of several investors, that VC will wish to have a large enough holding relative to the other major shareholders to be able to influence company policy.

## Interim Summary

This note began with a presentation of the basic valuation method, assuming no future dilution. The basic venture capital formula was derived, as well as formulas for computing the number and price of new shares. Any transaction at a fixed price for a fixed percent of outstanding shares implies a valuation for the entire company. The derivation, and the pitfalls, of implied valuations were discussed.

Formulas can all too easily imply the presence of objectivity and certainty. Each of the elements of the basic valuation formula embodies a number of assumptions. The choice of discount rate, the terminal value estimate, and the determination of the investment amount are each formed based on experience and subjective judgment. Although patterns can be detected, and rules of thumb justified in some cases, the experienced investor will also adopt his habits to suit the investment at hand, and its particular stage, industry, challenges, and pressures.

## Valuation Assuming Future Dilution

At the beginning of this note, MicroElectronics Enterprises was valued under the assumption that it would receive only one round of financing before its public offering in year five. As a rule, however, venture capital financing is spread out over a number of rounds, loosely corresponding to the various stages discussed above. As we have seen, financings at later stages command lower discount rates. Thus, staging capital commitments, rather than concentrating them in one early round, can work to create value for the founders of a venture.

Shares will be issued not only to satisfy capital needs, but also to provide incentives for key employees. It is in the VC's interest for managers to be motivated to maximize value. Thus, shares are normally reserved for management incentives at the earliest financing rounds.

As new stock is issued to later-round investors or to new key employees, the early-round investors can expect to suffer *dilution*, a loss of ownership due to the issuing of additional shares. As a result, the early-round investor will have to purchase a higher ownership percentage as of the financing, in order to achieve a given terminal or final ownership after future financings. However, if more stock is issued to early investors, the future investors will have to receive more stock to receive a given percent ownership. Thus, to determine the necessary current ownership, the VC must estimate the amount of new stock that will be issued in the future, but the amount of stock to be issued in the future depends in part on the amount of stock that will be issued in the current round. How can the VC solve this paradox?

## Summary of the Method

In order to avoid the circularity problem in which the current shares and the future shares each determine the other, the valuation procedure will be based on working only with final ownership

shares, the percent ownership of each party as of the terminal year. After first carving up the final year's pie, the final ownerships can then be converted to ownership levels as of each financing, and then into per-share prices.

1. *Estimate the size of the future pie.* How much value will be available to investors and management? The first step is to calculate the terminal value.
2. *Carve up the future pie.* Using the basic venture capital formula, the final ownership required by each investor can be estimated. First, the timing and amount of equity infusions must be projected, based on the venture's external financing needs. The total of the investors' final ownership amounts, plus the ownership to be allocated to future managers, must be less than 100%, with enough ownership left over for the founders. If not, then one of the parties will have to lower its expected return, or else the deal will not be feasible given the projections.
3. *Convert the future pie into the present.* This step is pure algebra. Given the ownership levels determined above, one can then determine the current ownerships, the number of new shares, and the share prices to establish as of each financing.

### *Estimating the Future Pie*

Estimating terminal value has been discussed in detail above. The procedure is the same here as it was in the example of valuation without dilution. Applying the price/earnings ratio approach to ME2, a PER of 15 times year-five earnings of \$2.5 million would provide a terminal valuation of \$37.5 million.

### *Carving up the Future Pie*

In our single-stage financing of ME2, the first round of financing was assumed to be sufficient to provide funds until the final year. A more realistic case will be the example where several rounds are required. A revised pricing worksheet for ME2, in which a total of three financing rounds are expected, is shown in **Exhibit 10**. In addition to estimating the appropriate discount rate for the current round, the first-round VC must now estimate the discount rates that are most likely to be applied in the following rounds, which are projected for years two and four, as summarized in **Exhibit 10**, part 3. Although a 50% rate is still appropriate for year 0, it is estimated that investors in this company will demand a 40% return in year 2, and a 25% return in year 4 (see **Exhibit 10**, part 6). The final ownership that each investor must be left with, given a terminal PER of 15, can be calculated using the basic valuation formula:

Round 1:

$$\text{Final \%} = \frac{\text{Future Value (Investment)}}{\text{Terminal Value (Company)}} = \frac{1.50^5 \times \$1.5\text{mm}}{15 \times \$2.5\text{mm}} = 30.4\%$$

$$\text{Round 2:} \quad (1.40^3 \times \$1.0\text{mm}) / (15 \times \$2.5\text{mm}) = 7.3\%$$

$$\text{Round 3:} \quad (1.25^1 \times \$1.0\text{mm}) / (15 \times \$2.5\text{mm}) = 3.3\%$$

Note that these are the final, or terminal, ownership shares that each investor requires. Because they are calculated from terminal value, current investment amount, and discount rate, they are not affected by the amount or pricing of future investments. That is, each investor's required final

ownership stake is not influenced by the other investors' final ownership percentages. However, because the early-round investor must purchase enough shares now to make up for the dilution that will be caused by the future financings, it is in the conversion from future to current ownership shares that the effect of future financings will be felt.

These required final ownership percents are shown in part 8 of **Exhibit 10**, for a range of possible terminal PERs. Also indicated is the portion of final ownership percentage to be granted future employees (which is 0% for the purposes of this example; if there were a final ownership share specified for future management, it would be treated as a fourth and final round). The total of these ownership percentages, for a PER of 15, is 41%, leaving a comfortable 59% of the future pie available for the founders (or open for negotiation). However, if the appropriate PER for this company is 5, then the total required final ownership share equals 123% for capital supplies; there is not enough terminal value to satisfy the needs of the investors, let alone preserve the incentives of the founders.

What if, upon research and reflection, the VC realizes that the appropriate PER is indeed 5? One temptation is to simply increase the projected PER or the projected profitability until the sum of the required final ownership share declines to a reasonable level. This amounts, of course, to self-delusion and refusal to confront a real problem. Eventually, future investors, taking a less sanguine view of the venture's prospects, will strike a harder bargain than this VC is willing to admit now. Alternately, managers faced with inadequate stock incentives will threaten to resign or, perhaps worse, their work will reflect their opinions of their compensation. Given the suggestion provided by this calculation that there may not be enough value to justify the investments needed, the VC must seriously consider the feasibility of the deal.

### *Conversion to Current Ownership and Prices*

The ratio of the percent ownership a given investor will hold as of the terminal year of a project—the final ownership share—to the percent ownership he holds as of his investment—the current ownership share—is that investor's *retention percent*:

$$\text{retention \%} = \frac{\text{final \% ownership}}{\text{current \% ownership}}$$

The reduction in ownership over time is caused by the sale of new shares to new investors or the award of additional options to management. To illustrate, an investor's retention ratio will be 75%, if a later investor purchases 25% of the company. In general:

$$\text{retention \%} = 1 - (\text{total of future final \% ownerships})$$

The retention percent can also be thought of as that portion of the final ownership available to the current investor. (The equivalence of the above two definitions of retention percent is demonstrated in **Appendix 2**.)

Thus, because the second- and third-round investors in ME2 will hold 7.3% and 3.3% of the final ownership, the first-round investor will only retain  $1 - (7.3\% + 3.3\%) = 89.3\%$  of his original holding. The second-round investor will retain  $1 - 3.3\% = 96.7\%$  of his investment, and the third-round investor will retain 100%, since there will be no subsequent investors through year 5, the terminal year.

So what percent ownership should each investor purchase at the time of the financing? The formula below is equivalent to the ratio above:

$$\text{current \% ownership} = \frac{\text{final \% ownership}}{\text{retention \%}}$$

so, for the first round: current % ownership = 30.4% / 89.3% = 34.0%

second round: current % ownership = 7.3% / 96.7% = 7.6%

third round: current % ownership = 3.3% / 100% = 3.3%

Using the formulas presented earlier, we can calculate the number of shares that each investor must purchase to obtain the above ownerships, assuming that there were 1,000,000 shares outstanding before the first round. Recall that:

$$\text{new shares} = \frac{\% \text{ ownership}}{1 - (\% \text{ ownership})} \times \text{old shares}$$

so, for the first round: new shares =  $\frac{34.0\%}{1 - 34.0\%} \times 1,000,000 = 515,055$

second round: new shares =  $\frac{7.6\%}{1 - 7.6\%} \times 1,515,055 = 124,077$

third round: new shares =  $\frac{3.3\%}{1 - 3.3\%} \times 1,639,131 = 56,522$

The price per share can also be calculated; recall that:

$$\text{share price} = \frac{\text{investment}}{\text{new shares}}$$

so, for the first round: share price = \$1.5mm / 515,055 = \$2.91

second round: share price = \$1.0mm / 124,077 = \$8.06

third round: share price = \$1.0mm / 56,522 = \$17.69

The terminal (or final-year) share price can also be calculated. There will be 1,639,131 + 56,522 = 1,695,653 shares outstanding after the third round. Thus, if the market value is 15 times the year-5 earnings of \$2.5mm, the price per share will be: (15 x \$2.5mm) / 1,695,653 = \$22.12.

To check that this pricing is consistent with the assumed discount rates, note the rate of growth from each VC price to the terminal price:

first round:  $(22.12 / 2.91)^{1/5} - 1 = 50\%$

second round:  $(22.12 / 8.06)^{1/3} - 1 = 40\%$

third round:  $(22.12 / 17.69)^{1/1} - 1 = 25\%$

Thus, this pricing schedule will yield the returns that are expected.

### *The Value of Staged Capital Commitment*

Compare the final outcome of the ME2 pricing scenario above with three financing stages (summarized in **Exhibit 10**). with the single-stage investment examined at the beginning of this note (and summarized in **Exhibit 11**). With three stages, the various VCs would demand a total of 41.0% of the final ownership, leaving 59.0% for the founders. With only one financing round, 29.1% would be available for the founders. The difference in the value of the founders' holdings, 29.9% of the \$37.5mm terminal valuation, is \$11.2 million. (This is also equal to the difference in the final-year share price,  $\$22.12 - \$10.92 = \$11.20$ , times the one million founders' shares.)

This extra value is accounted for by the *value of the option to abandon* (or in reality renegotiate) the venture, which the VCs have in the three-stage scenario but do not have if they commit all their capital at the beginning. There is always a chance that things will go wrong between years zero and two. If the VC commits only a portion of the total funds that will be necessary over the venture's life, he preserves the option not to invest further. If, on the other hand, the VC invests, in year zero, the entire capital that will be required, he has exposed the entire investment, including funds that won't be required until year four, to the risk of the earliest years. In that case, he must apply the high first-round discount rate to that investment. If the venture goes well, the VC will have made a much higher profit than he would with a three-stage commitment. This higher profit, which comes at the expense of the founders, compensates the VC for foregoing the option to abandon (or renegotiate) the venture should things go poorly.<sup>13</sup> Finally, note that deferring taking in investment capital enables a company to benefit from lower discount rates for all the reasons discussed earlier.

### *Shares for Future Management*

Venture capitalists generally insist that managers own stock in their companies, in order to provide the highest incentive to create long-term value. Since a company's initial management team generally consists of the firm's founders, the final ownership that constitutes their incentive can be readily determined. It is that portion of the final ownership that is not required by current and future investors.

Early-stage ventures generally do not come with complete management teams. Crucial positions remain unfilled (for example, a marketing executive usually is not recruited until the product is ready for scale production); additional managers will be required as the company grows and as some of the current managers leave; and key non-management employees might be granted stock incentives as well. Just as the early-round venture capitalist must forecast capital needs and the requirements of future investors in order to compensate for dilution from future financing, so he must forecast personnel needs and the incentive requirements of future managers in order to compensate for dilution from management stock incentives.

In the ME2 example, none of the final ownership was set aside for management (see **Exhibits 10** and **11**, parts 5 and 8). More realistically, a venture capitalist might have provided for, say, 15% of the final ownership to be allocated for future management incentives (incentives for managers other than the founders). If so, then the founders' final ownership, as well as the retention percent of each investor, would be reduced by that same 15%. The ME2 three-stage financing scenario is shown in

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<sup>13</sup> Note that even when the cash has been invested, the company can be liquidated at any stage (i.e., the option to abandon can be invoked). In reality, once the money is inside the company, there is a high likelihood of total loss.

**Exhibit 12**, as modified (in part 5) to show the effects of dilution from management incentives. Note that, for the same required rate of return, and thus the same final ownerships, the investors must now receive a higher current ownership share, and thus a lower stock price, in each round as compared to the scenario of **Exhibit 10**.

The distribution of final ownership for ME2, assuming three financing rounds and 15% equity for additional management, is shown below. The actual distribution of ownership as of their initial public offering is shown for selected venture-backed companies in **Exhibit 11**.

### *Examining the Valuation: Some Questions Venture Capitalists Often Ask*

As was discussed earlier in the introduction of this note, valuation is part art and part science. The objective is not to search for the “correct” answer, but rather to lessen the degree of uncertainty. Venture capitalists, or any other similar group of professionals, use experience to guide them to the right questions. They have witnessed many births and deaths of companies. They are familiar with what can go right and what can go wrong in such ventures.

In this regard, it is useful to introduce a set of questions that are often asked by professional venture capitalists when they are considering how to price a given deal. These questions are presented below, with references to the model first introduced as **Exhibit 10**. The questions are addressed in **Exhibit 14**.

*What if we (or the later investors) change our required discount rate?* Table 1 shows the final ownership share required by each round’s investors at various discount rates, calculated using the basic venture capital formula. The investor can tally up the appropriate ownership shares for a quick check of the robustness of the deal under different assumptions. For example, should the next two rounds be calculated with discount rates of 50% and 40%, rather than 40% and 25% as forecast, the total required ownership would increase by only 2.1%, equal to  $(9.0\% + 3.7\%) - (7.3\% + 3.3\%)$ . This is negligible, reflecting the relatively small future financing needs of ME2. For a more capital-intensive enterprise, the resulting future dilution would have a more meaningful effect on first-round price.

*What if we change our required discount rate or the PER assumption?* Given a different assumed terminal PER, or a different discount rate for the current round’s investors, what ownership would the VCs need? Table 2 shows the results, derived by recalculating **Exhibit 12**, Table 10, for different rates and PERs. Note, for example, that the deal struck with a 50% rate and a PER of 15 (40.9% current ownership), is similar to the deal that would be struck with a 60% rate, but assuming a PER of 20 (40.8% ownership). Table 3 shows the corresponding valuations. It is calculated by dividing the first-round investment by the ownership percentages in Table 2.

*What if a different valuation is proposed?* Although this note begins the pricing process with discount rates and terminal values, venture capital negotiations are often conducted in terms of valuations. Given a \$5 million valuation proposed by an entrepreneur, the VC can use Table 3 to translate that value into its possible underlying assumptions: a 50% return at a PER of 20, a 40% return at a PER of 15, a 27% return (or so) at a PER of 10, and so on. The VC can then argue the feasibility of these assumptions.

*What if a different price is proposed?* Some negotiations, on the other hand, are conducted in the language of share price. Table 4 corresponds to Tables 2 and 3, and allows the VC to translate a \$4.50 share price offer (to illustrate) into the discount rates and PERs listed above. Table 4 was calculated by applying the formula for the number of new shares to Table 2; the result is divided into the first-round investment.

*What if different discount rates are demanded in the first two rounds?* If the first-round VC were to determine that a different discount rate were appropriate for himself, then his forecast of the appropriate next-round rate would also probably change. For example, if ME2 suddenly produced a major technological breakthrough or a few long-term contracts with major corporations, the first-round VC might determine that a 40% discount rate was appropriate. Under that scenario, the next round VC would probably demand only 25% or 30%, in which case, as Table 5 indicates, the current ownership demanded would be roughly 28.2%. Table 5 was calculated in a manner similar to Table 2. Table 6 shows the corresponding share prices.

The above questions all treat the situation where the discount rate demanded by the VC affects the pricing of the stock, and thus the dilution affecting the other shareholders. The remaining questions assume that the number of shares to be issued is fixed, regardless of the investor's desired rate of return or PER expectations. This is similar to the situation faced by the investor in an initial public offering. That investor is presented with a price, and can only decide whether or not to invest. He might determine a maximum acceptable purchase price, but the price he settles on will not determine the overall terms of the financing.

*What if the company's performance varies from plan?* What will the VCs' return be if ME2 only achieves half the forecast net income? Table 7 shows the rate of return for each investor, assuming a purchase at the prices indicated in Table 4, and a sale at a price reflecting the adjusted net income and the PER forecast in **Exhibit 11**. If only half the forecast terminal value is available then the first-round VC will still get a healthy 30.6% return, whereas the next two investors will only get 11.1% and -37.5% respectively. More realistically, later-round investors would be aware that ME2 will not make its year-five projections, and will negotiate a lower price. The increased dilution from these rounds, plus the lower PER that will probably accompany the more modest year-five performance of ME2, will serve to more dramatically reduce the first-round VCs' return.

*What if the investor seeks a certain return?* What must the company's performance be if the investor is to receive a given return? This question is, in a sense, the reverse of the previous question. If each financing round is priced as projected, what is the income that ME2 must produce in year five for the first-round VC to realize a 40% return? Table 8 shows that a net income of \$1.77 million would produce that return, and would also yield a 25% return to the second-round (year two) investors. Again, it is assumed that whatever PER is forecast in the pricing in **Exhibit 12** will apply in year five.

*What if a different terminal PER applies?* Table 10 shows the return to each round's investors, given various PERs. For example, should ME2 meet its projections but only command a PER of 10 in year 5, the three groups of investors will realize returns of only 38.3%, 22.3%, and -16.7%. The negative return to the year-four investor is unlikely to actually occur, since conditions that would cause ME2 to realize a PER of 10 would probably be apparent by year four, and would depress ME2's price in that year.

*What if a different terminal PER applies and company performance differs from plan?* What will the return to the first-round investor be if both terminal PER and net income vary? This question is a combination of two previous questions. Table 11 combines Tables 7 and 10 and shows the rate of return (to the first round investor only) under a range of different assumptions. For example, if ME2 only reaches 75% of plan, and thus commands a PER of 10, the investor will still receive a return of 30.6%.

*What if we expect different performance?* What price should the investor pay if he forecasts a different net income, for a given required rate of return. Table 12 shows shares prices corresponding to different rates and net incomes. For example, if an investor expects the company to perform to plan, but only desires a 40% rate of return, he would be willing to buy at \$3.07. Note that this is different

from the \$3.68 figure given in Tables 4 and 6, because those tables assume that, if a higher price is paid, fewer shares will be issued. Tables 7 and following assume that the price to the investor does not affect the company's shares outstanding (for example, the investor could be contemplating purchasing another investor's stock).

*What if we demand a different rate of return, or expect a different PER?* What must the company's performance be to yield a given rate of return, assuming a given PER? For example, an investor who expects a terminal PER of only 10 will want to know what net income and sales will be required to yield the 40% rate he requires. Tables 13 and 14 show that ME2 would have to achieve a net income of \$2.66 million, corresponding to sales of \$13.6 million.

The questions above assumed that the investor was presented with a fixed price. The investor, by asking those questions, could then evaluate that price under different assumptions for company performance, terminal PER, or discount rate. The next question and its accompanying set of tables examine the attractiveness of the stock over a range of possible prices.

*What if the stock becomes available at a different price?* If the same stock were available at 50% above the expected price, would we still want to buy it? Table 15 shows that, if the stock were to sell at \$3.26, an investor would only receive a 38.3% rate of return, even if the terminal PER remained at 15. Tables 16, 17, and 18 show that, if the investor who buys at this price is to receive a 50% return, the stock would have to sell at \$24.74 in year five, corresponding to a total net income of \$3.75 million and sales of \$19.2 million.

The final two questions do not relate to current stock price. Rather, they are designed to assist in thinking about the venture's terminal value.

*What if we base the terminal value on capitalized cash flow?* Some investors are more accustomed to working with cash flows as a source of value rather than PERs. Capitalized cash flow might be appropriate for a corporation considering a long-term investment, as opposed to a private investor expecting to sell his investment in the public market. Table 19 converts combinations of growth rates and capitalization rates to the PERs used in the other tables. Terminal free cash flow is calculated using the growth rates in the table (the actual year-five growth rate is probably not sustainable, thus a long-term growth rate should be selected from the table), and is divided by (cap. rate) - (growth rate) to yield a present value, which is divided by the terminal net income to yield a price/earnings ratio.

### *Summary and Conclusion*

The objective of this note was to present a method for valuing high risk, long-term investments. In the context of the Entrepreneurial Finance course, the objective is to present a way of translating future results into current, actionable values. In this sense, the characterization of the content of this note as a "method" is somewhat too ambitious: what has been presented is a way of thinking about a broad class of interesting investments.

The thought process entails certain important steps:

1. Forecasting future results for some point in the future;
2. Determining a value that is likely to apply at that point, based on well-understood valuation methodologies (e.g., price earnings ratios for comparable publicly traded companies);

3. Determining how much capital will be required to attain the target level of sales as well as the likely timing of stock sales;
4. Determining the amount of new shares that will have to be awarded to attract new managers required to attain the targets;
5. Determining the share of the final value “pie” that each group will demand, given a set of required rates of return, where the discount rates depend on:
  - a) The level of risk-free interest rates in the economy;
  - b) A premium for risk;
  - c) A return premium for illiquidity;
  - d) A premium for value-added by the investor; and,
  - e) A premium related to past experience using forecasted results;
6. Converting the future values back to the present in order to determine share prices and ownership percentages that make sense; and, finally,
7. Engaging a thought process, based on experience, to ask a set of questions that relate to what can go wrong and what can go right for the venture and valuation levels in the capital markets, as well as a set of questions designed to shed light on the reasonableness of the projections.

While many venture capitalists rely to some degree on some form of the “method” described in this note, many more would probably actively decry efforts to create an analytical model of the valuation process. They would point out that errors in valuation are not generally the cause of failure in ventures. The key determinant of success is forming partnerships with the right people who are pursuing the right opportunity.

The reader must keep these criticisms or caveats in mind. A sharper pencil will not necessarily make you a better evaluator of investment decisions. However, sharper pencils have their role, and some ventures would benefit from better analytics.

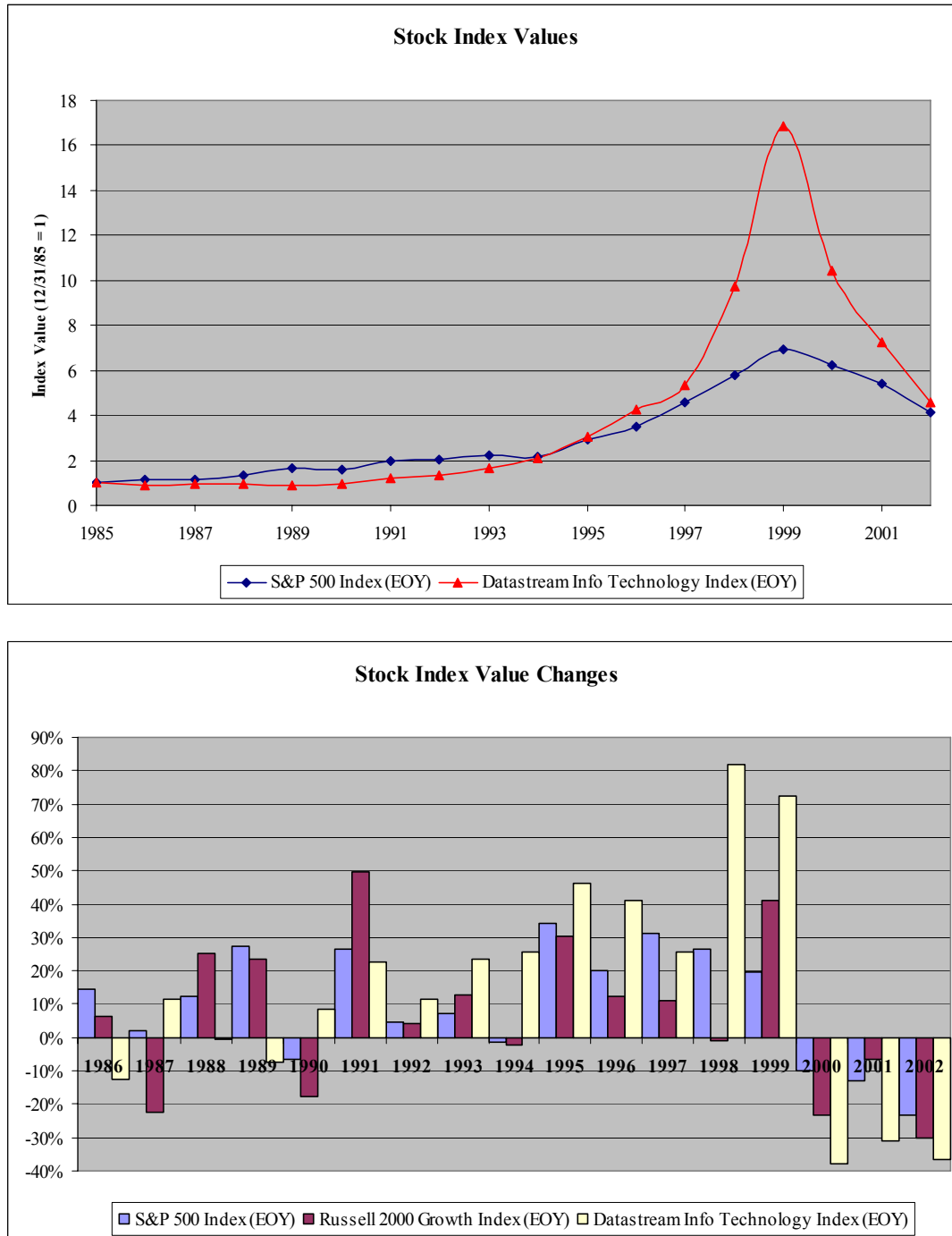
Moreover, in a world dominated with “rules of thumb,” there will often be opportunities to those who engage their brains rather than relying on simplistic rules. This is so because these rules of thumb do not always change when conditions change. A constant theme of Entrepreneurial Finance is that change is a source of opportunity, and this is an area where the theme holds true. Negotiating against those who cling to outdated rules of thumb is often a source of great advantage.

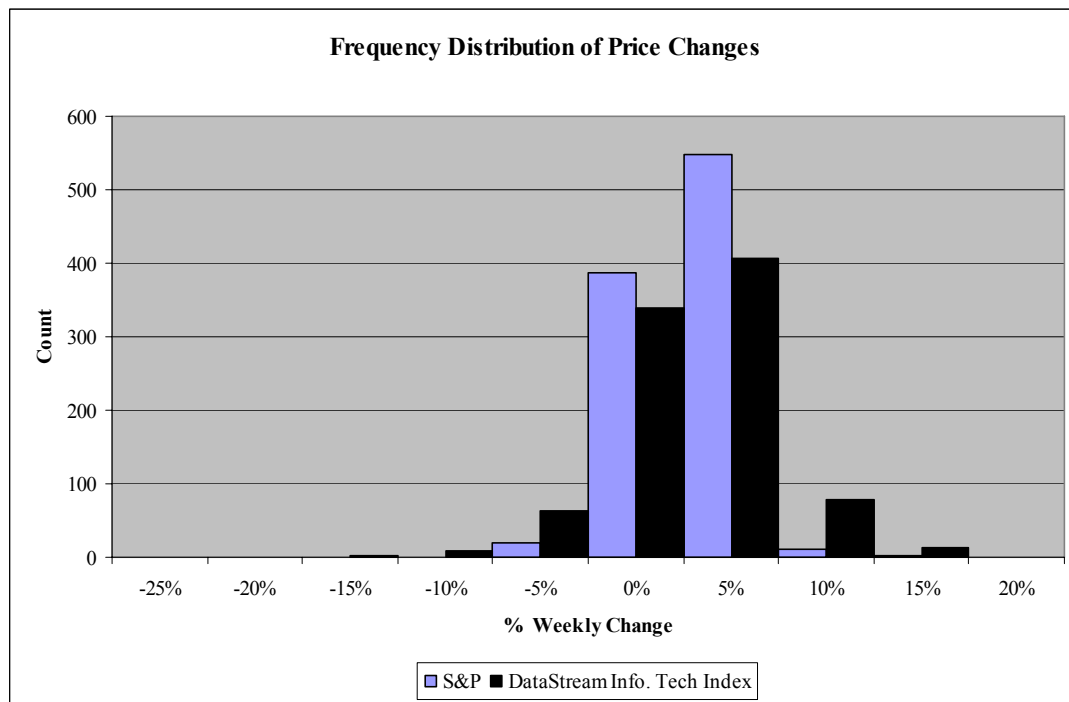
In closing, valuing high risk, long-term investments is intellectually challenging. However, decisions must be made and the consequences endured. The methodology described in this note has proven useful to those charged with the responsibility of making fundamental valuation decisions in this area. By utilizing the underlying thought process, tempered and honed by experience, you will be able to make more sensible decisions, those that improve the potential reward to risk ratio.

**Exhibit 1** Selected Data on Historical Asset Returns

| Series                                   | Geometric Mean<br>(1926-2002) | Arithmetic Mean<br>(1926-2002) | Standard Deviation<br>(1926-2002) |
|------------------------------------------|-------------------------------|--------------------------------|-----------------------------------|
| Basic Series: Nominal Returns            |                               |                                |                                   |
| Large Company Stocks                     | 10.2%                         | 12.2%                          | 20.5%                             |
| Small Company Stocks                     | 12.1%                         | 16.9%                          | 33.2%                             |
| Long-Term Corporate Bonds                | 5.9%                          | 6.2%                           | 8.7%                              |
| Long-Term Government Bonds               | 5.5%                          | 5.8%                           | 9.4%                              |
| Intermediate-Term Government Bonds       | 5.4%                          | 5.6%                           | 5.8%                              |
| U.S. Treasury Bills                      | 3.8%                          | 3.8%                           | 3.2%                              |
| Inflation                                | 3.0%                          | 3.1%                           | 4.4%                              |
| Basic Series: Real Returns               |                               |                                |                                   |
| Large Company Stocks                     | 6.9%                          | 9.0%                           | 20.6%                             |
| Small Company Stocks                     | 8.8%                          | 13.1%                          | 32.5%                             |
| Long-Term Corporate Bonds                | 2.8%                          | 3.0%                           | 10.2%                             |
| Long-Term Government Bonds               | 2.3%                          | 2.7%                           | 10.4%                             |
| Intermediate-Term Government Bonds       | 2.3%                          | 2.6%                           | 7.0%                              |
| U.S. Treasury Bills                      | 0.2%                          | 0.8%                           | 4.0%                              |
| Derived Series                           |                               |                                |                                   |
| Equity Risk Premia (Stocks-Bills)        | 6.2%                          | 8.2%                           | 20.2%                             |
| Small Stock Premia (Small Stocks-Stocks) | 1.8%                          | 3.4%                           | 18.4%                             |
| Default Premia (L/T Corps - L/T Gov't)   | 0.4%                          | 0.5%                           | 3.1%                              |
| Horizon Premia (LT Gov't - Bills)        | 1.6%                          | 2.0%                           | 8.7%                              |
| Real Interest Rates (Bills - Inflation)  | 0.7%                          | 0.7%                           | 4.2%                              |

Source: Ibbotson Associates, Stocks, Bonds, Bills, and Inflation—2003 Yearbook, Ibbotson Associates, Chicago, 1987, pp. 33, 39, 87-89, 97-99.

**Exhibit 2** Selected Data on Stock Indices

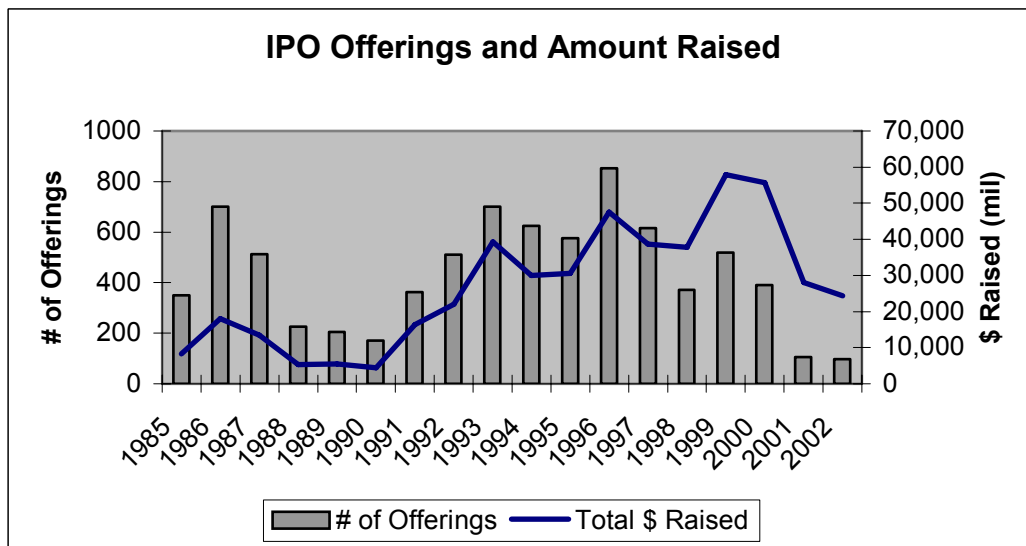
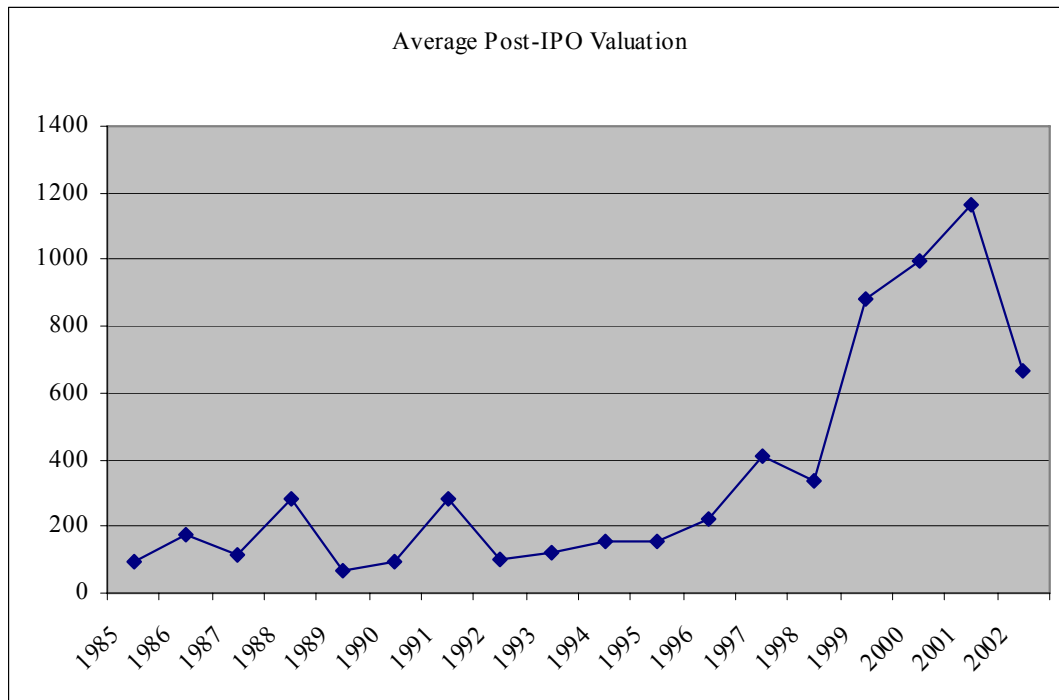
**Exhibit 2** Selected Data on Stock Indices (Continued)

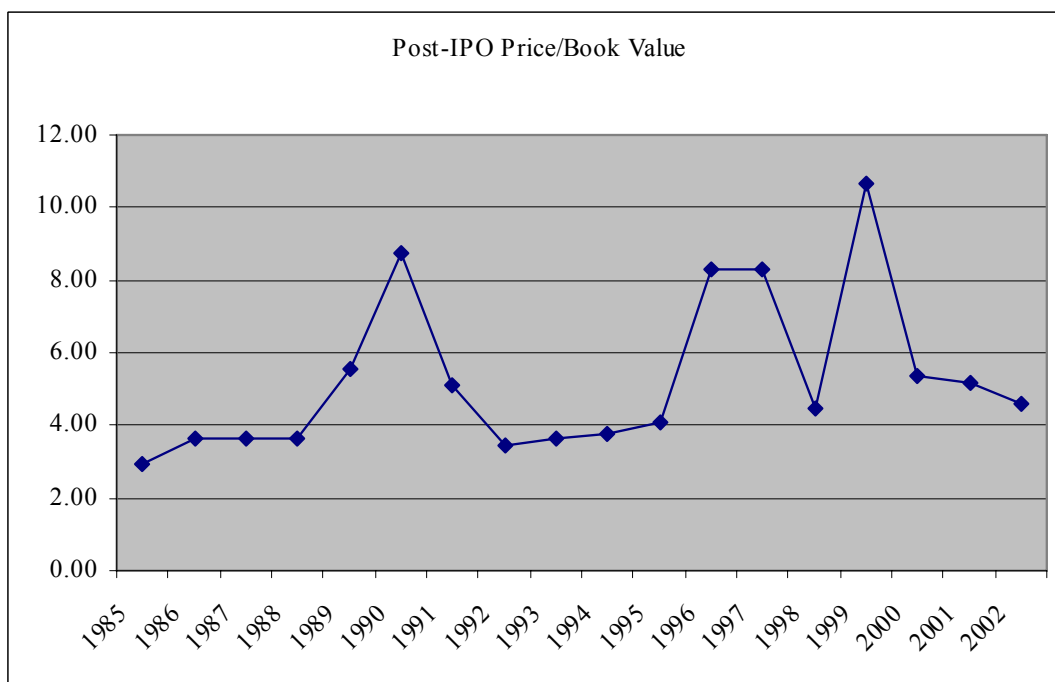
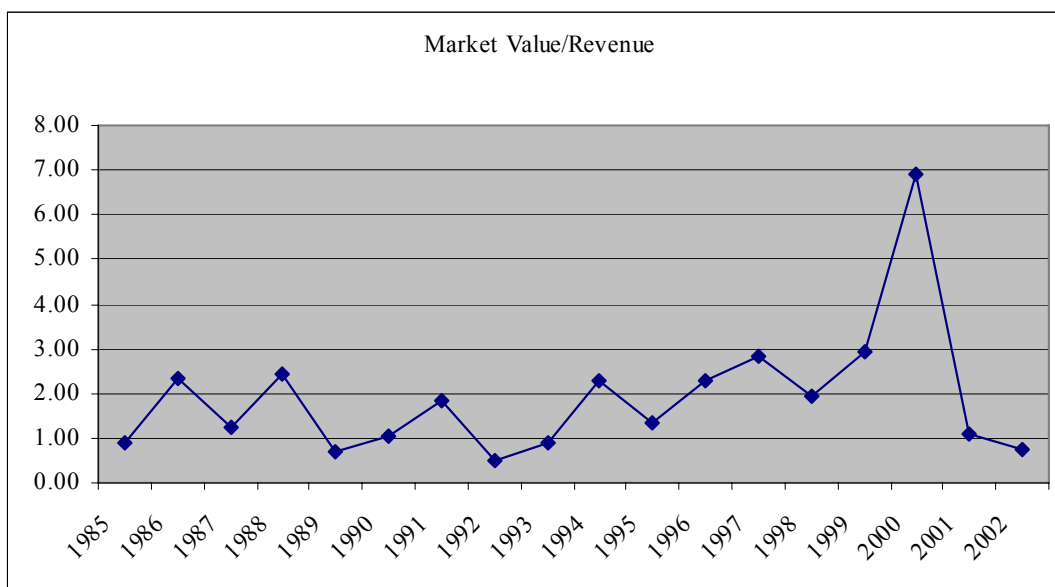
Source: DataStream.

## Exhibit 3 Selected Macroeconomic and Financial Data

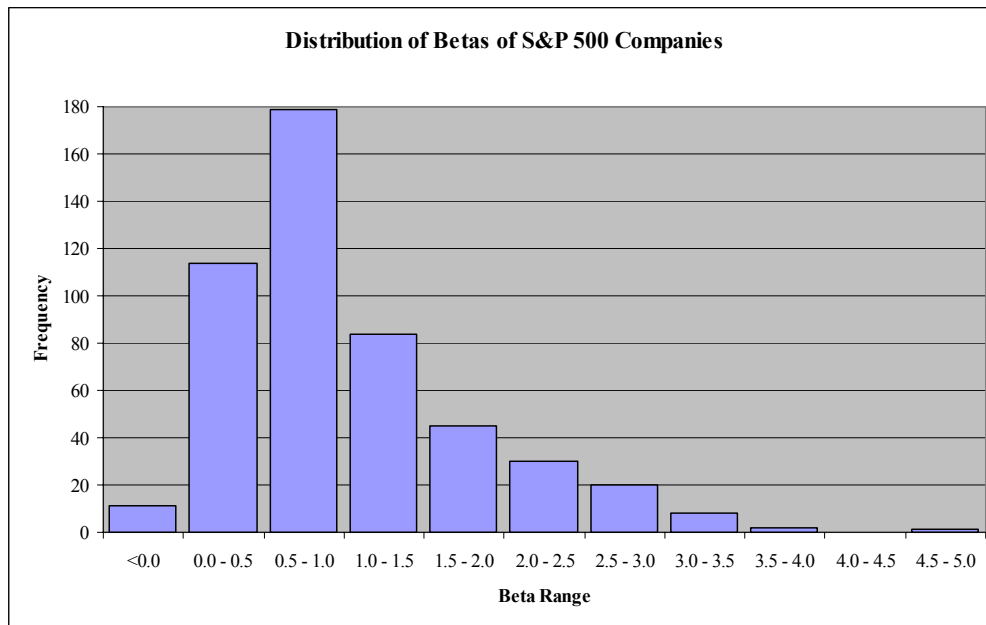
|                                                   | 1986   | 1987   | 1988  | 1989  | 1990  | 1991   | 1992   | 1993   | 1994   | 1995   | 1996   | 1997   | 1998   | 1999   | 2000    | 2001   | 2002   |
|---------------------------------------------------|--------|--------|-------|-------|-------|--------|--------|--------|--------|--------|--------|--------|--------|--------|---------|--------|--------|
| <b>ECONOMIC INDICATORS</b>                        |        |        |       |       |       |        |        |        |        |        |        |        |        |        |         |        |        |
| 1 Gross domestic product                          | 4,453  | 4,743  | 5,108 | 5,489 | 5,803 | 5,986  | 6,319  | 6,642  | 7,054  | 7,401  | 7,813  | 8,318  | 8,782  | 9,274  | 9,825   | 10,082 | 10,446 |
| 2 % chg in Real GDP                               | 3.4    | 0.07   | 4.2   | 3.5   | 1.8   | -0.5   | 3.0    | 2.7    | 4.0    | 2.7    | 3.6    | 4.4    | 4.3    | 4.1    | 3.8     | 0.3    | 2.4    |
| 3 CPI                                             | 109.6  | 113.6  | 118.3 | 124.0 | 130.7 | 136.2  | 140.3  | 144.5  | 148.2  | 152.4  | 156.9  | 160.5  | 163.0  | 166.6  | 172.2   | 177.1  | 179.9  |
| 4 Unemployment                                    | 7.0    | 6.2    | 5.5   | 5.3   | 5.6   | 6.8    | 7.5    | 6.9    | 6.1    | 5.6    | 5.4    | 4.9    | 4.5    | 4.2    | 4.0     | 4.8    | 5.8    |
| 5 Total Industrial Production Index               | 67.1   | 70.7   | 74.3  | 74.8  | 75.2  | 73.7   | 76.3   | 78.9   | 83.7   | 88.1   | 92.2   | 100.0  | 106.5  | 111.8  | 117.4   | 112.6  | 111.5  |
| 6 New Business Incorporations (000s)              | 703    | 686    | 685   | 677   | 647   | 629    | 667    | 707    | 742    | 767    | 786    | 799    | na     | na     | na      | na     | na     |
| 7 Corporate profits (\$bn) *                      | 301    | 347    | 405   | 396   | 409   | 431    | 453    | 511    | 573    | 669    | 754    | 834    | 777    | 806    | 788     | 731.6  | na     |
| 8 S&P 500 Dividend Yield                          | 3.49   | 3.08   | 3.64  | 3.45  | 3.61  | 3.24   | 2.99   | 2.78   | 2.82   | 2.56   | 2.19   | 1.77   | 1.49   | 1.25   | 1.15    | 1.32   | 1.61   |
| 9 S&P 500 P/E Ratio                               | 13.78  | 16.34  | 15.64 | 12.19 | 14.71 | 15.21  | 23.3   | 24.31  | 23.05  | 16.86  | 17.47  | 20.58  | 23.88  | 32.34  | 33.29   | 24.73  | 40.6   |
| <b>STOCK, BOND, AND VC INDICIES AND RETURNS</b>   |        |        |       |       |       |        |        |        |        |        |        |        |        |        |         |        |        |
| 10 VE Post Venture Capital Index (EOY)            | 96     | 97     | 107   | 133   | 123   | 221    | 236    | 269    | 281    | 408    | 472    | 484    | 706    | 1,360  | 764     | 543    | 323    |
| 11 S&P 500 Index (EOY)                            | 242    | 247    | 278   | 353   | 330   | 417    | 436    | 466    | 459    | 616    | 741    | 970    | 1,229  | 1,469  | 1,320   | 1,148  | 880    |
| 12 Russell 2000 Growth Index (EOY)                | 632    | 490    | 613   | 756   | 623   | 932    | 970    | 1,093  | 1,068  | 1,392  | 1,562  | 1,735  | 1,721  | 2,430  | 1,869   | 1,747  | 1,223  |
| 13 Datastream Info Technology Index (EOY)         | 144    | 160    | 159   | 147   | 160   | 196    | 219    | 270    | 340    | 497    | 701    | 881    | 1,603  | 2,765  | 1,716   | 1,187  | 751    |
| 14 Datastream U.S. Real Estate Index (EOY)        | 407    | 392    | 428   | 485   | 276   | 401    | 451    | 596    | 563    | 667    | 919    | 1,181  | 904    | 784    | 954     | 971    | 933    |
| 15 MSCI World (ex US) Stock Index (EOY)           | 591    | 725    | 912   | 1,002 | 757   | 834    | 717    | 933    | 987    | 1,081  | 1,138  | 1,146  | 1,341  | 1,706  | 1,449   | 1,122  | 927    |
| 16 Large Company Stocks Total Return (%)          | 18.5   | 5.2    | 16.8  | 31.5  | -3.2  | 30.6   | 7.7    | 10.0   | 1.3    | 37.4   | 23.1   | 33.4   | 28.6   | 21.0   | -9.1    | -11.88 | -22.1  |
| 17 Small Company Stocks Total Returns (%)         | 6.9    | -9.3   | 22.9  | 10.2  | -21.6 | 44.6   | 23.4   | 21.0   | 3.1    | 34.5   | 17.6   | 22.8   | -7.3   | 29.8   | -3.6    | 22.77  | -13.28 |
| 18 Long-Term Government Bonds Total Returns (%)   | 24.5   | -2.7   | 9.7   | 18.1  | 6.2   | 19.3   | 8.1    | 18.2   | -7.8   | 31.7   | -0.9   | 15.9   | 13.1   | -9.0   | 21.5    | 3.7    | 17.84  |
| 19 U.S. Treasury Bills Total Returns (%)          | 6.2    | 5.5    | 6.4   | 8.4   | 7.8   | 5.6    | 3.5    | 2.9    | 3.9    | 5.6    | 5.2    | 5.3    | 4.9    | 4.7    | 5.9     | 3.83   | 1.65   |
| 20 All PE Backed IPO 1-Year Total Returns (%)     | 4.5    | 49.0   | 21.8  | 22.8  | 20.8  | 19.2   | 63.2   | 35.8   | 28.4   | 95.1   | 63.2   | 69.5   | 71.5   | 61.3   | 62.2    | 55.5   | 39.9   |
| <b>VENTURE CAPITAL AND IPO ACTIVITY</b>           |        |        |       |       |       |        |        |        |        |        |        |        |        |        |         |        |        |
| 21 Number of Funds Raised                         | 133    | 142    | 141   | 160   | 140   | 76     | 124    | 166    | 201    | 267    | 289    | 451    | 569    | 846    | 1,442   | 707    | 386    |
| 22 Net Period Amount Raised (\$mil)               | 4,560  | 5,258  | 5,839 | 7,247 | 4,786 | 3,244  | 6,837  | 9,322  | 12,259 | 14,181 | 16,861 | 29,532 | 43,464 | 80,363 | 158,341 | 63,789 | 10,490 |
| 23 Estimated Disbursements to Portfolio Companies | 2,900  | na     | na    | na    | na    | na     | na     | 3,794  | 4,148  | 7,638  | 11,610 | 15,080 | 21,448 | 54,794 | 106,270 | 40,825 | 21,279 |
| <b>IPO DATA</b>                                   |        |        |       |       |       |        |        |        |        |        |        |        |        |        |         |        |        |
| 24 Number of Offerings                            | 701    | 513    | 225   | 205   | 171   | 363    | 510    | 700    | 624    | 576    | 852    | 616    | 372    | 520    | 391     | 106    | 97     |
| 25 IPO Amount (\$mil)                             | 18,062 | 13,503 | 5,272 | 5,424 | 4,383 | 16,359 | 22,003 | 39,327 | 29,920 | 30,573 | 47,520 | 38,629 | 37,828 | 57,912 | 55,740  | 28,118 | 24,395 |
| 26 # of Offerings (VC Backed Co's)                | 188    | 65     | 28    | 33    | 35    | 79     | 71     | 93     | 80     | 97     | 130    | 65     | 36     | 123    | 107     | 21     | 12     |
| 27 IPO Amount (\$mil)                             | 2,014  | 1,354  | 470   | 520   | 706   | 2,739  | 3,251  | 2,339  | 2,103  | 3,676  | 5,226  | 2,640  | 1,863  | 9,966  | 11,409  | 1,801  | 1,686  |

Source: 1-7, 2003 Economic Report to the President; 8-9, DataStream; 10, VentureXpert; 11-14, Datastream; 15, MSCI; 16-19, Ibbotson Associates; 20-23, VentureXpert; 24-25, SBC Platinum; 26-27, VentureXpert.

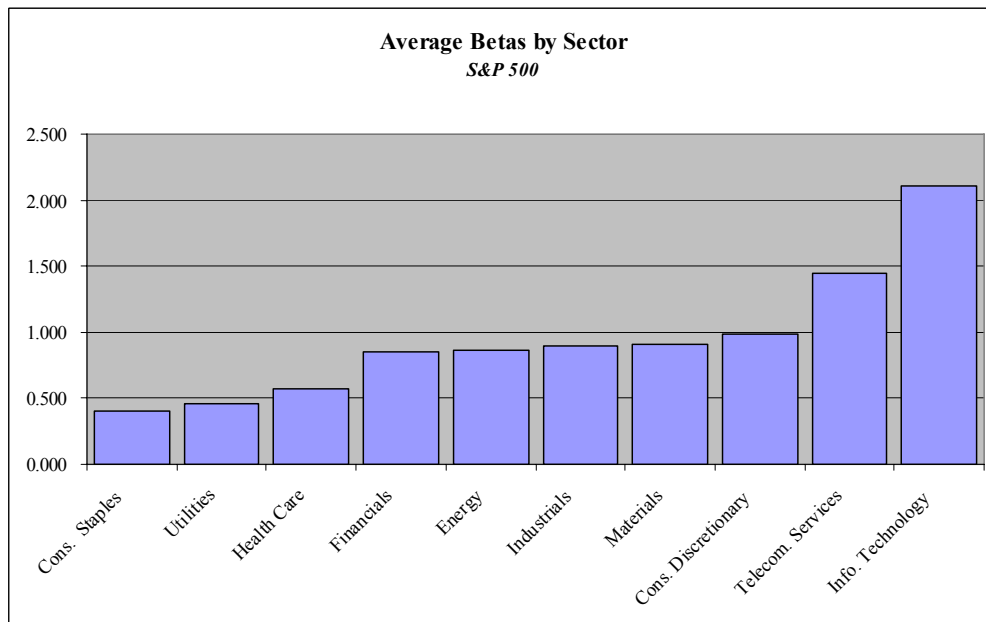
**Exhibit 4** Historical Valuation Trends in Initial Public Offerings

**Exhibit 4** Historical Valuation Trends in Initial Public Offerings (continued)

Source: SBC Platinum.

**Exhibit 5** Selected Data on Distribution of Systematic Risk Levels: 1998-2002

Source: DataStream.

**Exhibit 6** Selected Data on Systematic Risk Levels By Sector: 1998-2002

Source: DataStream.

**Exhibit 7** Betas for Selected Public Companies

| <b>Industry</b>              | <b>Company Name</b>          | <b>Beta</b> | <b>Market Value (\$mil)</b> |
|------------------------------|------------------------------|-------------|-----------------------------|
| Biotechnology                | IMMUNOGEN INC                | 2.0         | 163                         |
| Biotechnology                | MILLENNIUM PHARMACTCLS INC   | 1.9         | 4,620                       |
| Computers & Peripherals      | INTEL CORP                   | 1.9         | 136,038                     |
| Computers & Peripherals      | DELL COMPUTER CORP           | 1.7         | 80,562                      |
| Computers & Peripherals      | SUN MICROSYSTEMS INC         | 2.6         | 13,898                      |
| Diversified Financials       | CITIGROUP INC                | 1.5         | 211,173                     |
| Diversified Telecomm Svcs    | AT&T CORP                    | 0.9         | 15,311                      |
| Food & Drug Retailing        | WAL-MART STORES              | 0.9         | 230,731                     |
| Household Products           | PROCTER & GAMBLE CO          | 0.0         | 118,880                     |
| Industrial Conglomerates     | GENERAL ELECTRIC CO          | 1.1         | 286,838                     |
| Internet & Catalog Retail    | AMAZON.COM INC               | 2.8         | 14,059                      |
| Internet & Catalog Retail    | EBAY INC                     | 2.6         | 32,274                      |
| Internet & Catalog Retail    | USA INTERACTIVE              | 1.6         | 17,495                      |
| Internet Software & Services | YAHOO INC                    | 3.4         | 18,000                      |
| IT Consulting & Services     | ELECTRONIC DATA SYSTEMS CORP | 1.2         | 9,639                       |
| Media                        | AOL TIME WARNER INC          | 2.7         | 65,841                      |
| Oil & Gas                    | EXXON MOBIL CORP             | 0.4         | 243,130                     |
| Pharmaceuticals              | JOHNSON & JOHNSON            | 0.4         | 161,355                     |
| Pharmaceuticals              | MERCK & CO                   | 0.4         | 124,629                     |
| Software                     | INTUIT INC                   | 1.7         | 9,446                       |
| Software                     | MICROSOFT CORP               | 1.7         | 263,361                     |
| Tobacco                      | ALTRIA GROUP INC             | 0.3         | 83,626                      |

Source: DataStream.

## Exhibit 8 Selected Public Internet Firms

| Company Name              | Market Value |      | Sales (\$MM) |      | EBITDA (\$MM) |      | Net Income |      | Total Assets |      | Price/Earnings |           | MV/Sales |       | MV/Assets |       |
|---------------------------|--------------|------|--------------|------|---------------|------|------------|------|--------------|------|----------------|-----------|----------|-------|-----------|-------|
|                           | (\$MM)       | FY02 | FY02         | FY02 | FY02          | FY02 | (\$MM)     | FY02 | (\$MM)       | FY02 | Ratio (\$MM)   | 7/25/2003 | Ratio    | Ratio | Ratio     | Ratio |
| ADOBE SYSTEMS INC.        | 6,850        |      | 1,165        |      | 382           |      | 191        |      | 1,052        |      | 36             |           | 6        |       | 7         |       |
| AMAZON.COM INC            | 7,328        |      | 3,933        |      | 193           |      | (150)      |      | 1,990        |      | (101)          |           | 2        |       | 4         |       |
| EBAY INC                  | 21,111       |      | 1,214        |      | 431           |      | 250        |      | 4,124        |      | 101            |           | 17       |       | 5         |       |
| ELECTRONIC ARTS INC       | 8,465        |      | 2,482        |      | 611           |      | 317        |      | 2,360        |      | 32             |           | 3        |       | 4         |       |
| EXPEDIA INC               | 3,946        |      | 591          |      | 159           |      | 70         |      | 860          |      | 100            |           | 7        |       | 5         |       |
| INFOSYS TECHNOLOGIES -ADR | 8,128        |      | 754          |      | 258           |      | 195        |      | 704          |      | 36             |           | 11       |       | 12        |       |
| INTERACTIVECORP           | 10,322       |      | 4,621        |      | 538           |      | 7          |      | 15,663       |      | 49             |           | 2        |       | 1         |       |
| MACROMEDIA INC            | 739          |      | 337          |      | 46            |      | 2          |      | 527          |      | 700            |           | 2        |       | 1         |       |
| MICROSOFT CORP            | 293,137      |      | 28,365       |      | 13,584        |      | 7,829      |      | 67,646       |      | 29             |           | 10       |       | 4         |       |
| ORACLE CORP               | 43,014       |      | 9,673        |      | 3,934         |      | 2,224      |      | 10,800       |      | 27             |           | 4        |       | 4         |       |
| OVERTURE SERVICES INC     | 1,618        |      | 668          |      | 121           |      | 73         |      | 436          |      | 19             |           | 2        |       | 4         |       |
| YAHOO INC                 | 9,726        |      | 953          |      | 198           |      | 107        |      | 2,790        |      | 136            |           | 10       |       | 3         |       |

Source: Standard &amp; Poor's.

## Exhibit 9 Selected Valuation and Performance Data on Companies

| Company Name                   | Industry                     | Price -<br>7/24/03 | 2001 Basic EPS<br>(Includ Extra) | 2002 Basic EPS<br>(Includ Extra) | % chg | P/E Ratio | 1 Year Total<br>Return (%) | 5 Year Total<br>Return (%) | Revenue<br>(\$mil) | Net Income<br>(\$mil) | Total Assets<br>(\$mil) | Market<br>Value (\$mil) | Avg. Daily<br>Shares<br>Traded | Net Profit<br>Margin (%) | Market/<br>Book Value<br>Ratio | Market<br>Value/<br>Revenue<br>Ratio |
|--------------------------------|------------------------------|--------------------|----------------------------------|----------------------------------|-------|-----------|----------------------------|----------------------------|--------------------|-----------------------|-------------------------|-------------------------|--------------------------------|--------------------------|--------------------------------|--------------------------------------|
| IMMUNOGEN INC                  | Biotechnology                | 4.01               | (0.42)                           | (0.37)                           | na    | (9.5)     | (1.0)                      | 17.2                       | 6                  | (15)                  | 132                     | 163                     | 142                            | na                       | 1.5                            | 27.7                                 |
| MILLENNIUM PHARMACEUTICALS INC | Biotechnology                | 15.55              | (0.88)                           | (2.13)                           | na    | na        | na                         | 28.7                       | 353                | (290)                 | 3,998                   | 4,620                   | 7,276                          | na                       | na                             | 1.6                                  |
| 3COM CORP                      | Communications Equipment     | 4.91               | (2.80)                           | (1.71)                           | na    | na        | (11.7)                     | (1.1)                      | 1,478              | (296)                 | 2,237                   | 1,779                   | 4,023                          | na                       | 1.0                            | 1.2                                  |
| CISCO SYSTEMS INC              | Communications Equipment     | 16.41              | (0.14)                           | 0.26                             | na    | 34.9      | 4.0                        | 5.4                        | 18,915             | 1,893                 | 37,795                  | 115,178                 | 63,625                         | 10                       | 4.2                            | 6.1                                  |
| APPLE COMPUTER INC             | Computers & Peripherals      | 17.95              | (0.07)                           | 0.18                             | na    | na        | (23.0)                     | 6.2                        | 5,742              | 65                    | 6,298                   | 6,562                   | 9,206                          | 1                        | 1.6                            | 1.1                                  |
| DELL COMPUTER CORP             | Computers & Peripherals      | 31.37              | 0.84                             | 0.48                             | -43%  | 36.1      | 16.8                       | 8.8                        | 33,404             | 2,122                 | 15,470                  | 80,562                  | 23,896                         | 6                        | 15.9                           | 2.3                                  |
| HEWLETT-PACKARD CO             | Computers & Peripherals      | 19.30              | 0.21                             | (0.36)                           | na    | na        | 4.4                        | (3.6)                      | 56,588             | (923)                 | 70,710                  | 39,523                  | 12,955                         | na                       | 1.6                            | 1.1                                  |
| INTEL CORP                     | Computers & Peripherals      | 20.82              | 0.19                             | 0.47                             | 147%  | 44.3      | (24.3)                     | 3.4                        | 26,764             | 3,117                 | 44,224                  | 136,038                 | 61,321                         | 12                       | 3.8                            | 5.1                                  |
| INTL BUSINESS MACHINES CORP    | Computers & Peripherals      | 88.04              | 4.45                             | 2.10                             | -53%  | 28.2      | 10.3                       | 9.0                        | 81,186             | 5,334                 | 96,484                  | 152,133                 | 7,123                          | 7                        | 6.2                            | 1.9                                  |
| SUN MICROSYSTEMS INC           | Computers & Peripherals      | 4.35               | 0.28                             | (0.18)                           | na    | na        | (36.9)                     | (2.8)                      | 12,496             | (387)                 | 16,322                  | 13,898                  | 63,373                         | na                       | 1.9                            | 1.1                                  |
| TERADYNE INC                   | Computers & Peripherals      | 17.15              | (1.15)                           | (3.93)                           | na    | na        | (36.7)                     | 2.2                        | 1,222              | (718)                 | 1,895                   | 3,355                   | 3,172                          | na                       | 3.3                            | 2.6                                  |
| AMERICAN EXPRESS               | Diversified Financials       | 41.66              | 0.99                             | 2.02                             | 104%  | 20.0      | (1.1)                      | 4.8                        | 24,649             | 2,671                 | 157,233                 | 54,031                  | 5,521                          | 11                       | 3.8                            | 2.2                                  |
| BANK OF AMERICA CORP           | Diversified Financials       | 74.20              | 4.26                             | 6.08                             | 43%   | 11.8      | 1.5                        | 3.2                        | 46,362             | 9,249                 | 660,438                 | 111,117                 | 5,144                          | 20                       | 2.2                            | 2.4                                  |
| CITICORP INC                   | Diversified Financials       | 41.02              | 2.79                             | 2.99                             | 7%    | 13.4      | 4.1                        | 9.0                        | 92,029             | 13,448                | 1,097,190               | 211,173                 | 14,326                         | 15                       | 2.4                            | 2.3                                  |
| AT&T CORP                      | Diversified Telecom Svcs     | 19.49              | 12.50                            | (17.53)                          | na    | 14.7      | (16.1)                     | (19.6)                     | 37,827             | 963                   | 55,272                  | 15,311                  | 6,560                          | 3                        | 1.2                            | 0.4                                  |
| VERIZON COMMUNICATIONS         | Diversified Telecom Svcs     | 37.85              | 0.14                             | 1.49                             | 964%  | 12.3      | (8.2)                      | (0.7)                      | 67,625             | 4,384                 | 167,468                 | 104,097                 | 6,088                          | 7                        | 2.9                            | 1.5                                  |
| WAL-MART STORES                | Food & Drug Retailing        | 52.61              | 1.41                             | 1.49                             | 6%    | 28.3      | (2.2)                      | 14.4                       | 244,524            | 8,039                 | 94,683                  | 230,731                 | 9,083                          | 3                        | 5.8                            | 0.9                                  |
| HUMANA INC                     | Health Care Providers & Svcs | 12.99              | 0.71                             | 0.87                             | 23%   | 11.4      | (14.7)                     | (16.0)                     | 11,281             | 143                   | 4,600                   | 2,098                   | 942                            | 1                        | 1.3                            | 0.2                                  |
| PROCTER & GAMBLE CO            | Household Products           | 91.82              | 2.15                             | 3.26                             | 52%   | 21.4      | 5.3                        | 3.8                        | 40,169             | 4,332                 | 40,776                  | 118,880                 | 3,241                          | 11                       | 7.8                            | 3.0                                  |
| GENERAL ELECTRIC CO            | Industrial Conglomerates     | 28.70              | 1.38                             | 1.42                             | 3%    | 18.3      | (5.1)                      | 2.3                        | 130,685            | 15,133                | 573,244                 | 286,838                 | 18,348                         | 12                       | 4.3                            | 2.2                                  |
| AMAZON.COM INC                 | Internet & Catalog Retail    | 35.89              | (1.56)                           | (0.39)                           | na    | na        | 96.9                       | 37.3                       | 3,933              | (150)                 | 1,990                   | 14,059                  | 9,103                          | na                       | (10.9)                         | 3.6                                  |
| EBAY INC                       | Internet & Catalog Retail    | 101.65             | 0.34                             | 0.87                             | 156%  | 98.7      | 84.1                       | na                         | 1,214              | 250                   | 41,124                  | 32,274                  | 6,525                          | 21                       | 8.2                            | 26.6                                 |
| USA INTERACTIVE                | Internet & Catalog Retail    | 38.24              | 1.03                             | 4.55                             | 342%  | 47.8      | 34.2                       | 25.6                       | 4,621              | 7                     | 15,663                  | 17,495                  | 12,154                         | 0                        | 2.2                            | 3.8                                  |
| EXPEDIA INC                    | Internet Software & Services | 73.28              | (0.83)                           | 0.59                             | na    | 96.4      | 105.0                      | na                         | 591                | 66                    | 860                     | 3,724                   | 2,732                          | 11                       | 16.6                           | 6.3                                  |
| YAHOO INC                      | Internet Software & Services | 29.84              | (0.16)                           | 0.07                             | na    | 124.3     | 86.3                       | 16.9                       | 953                | 107                   | 2,790                   | 18,000                  | 12,044                         | 11                       | 7.6                            | 18.9                                 |
| AMERICAN MANAGEMENT SYSTEMS    | IT Consulting & Services     | 13.62              | 0.38                             | 0.67                             | 76%   | 20.3      | (34.8)                     | (12.9)                     | 987                | 28                    | 622                     | 575                     | 116                            | 3                        | 1.3                            | 0.6                                  |
| ELECTRONIC DATA SYSTEMS CORP   | IT Consulting & Services     | 20.15              | 2.90                             | 2.33                             | -20%  | 17.2      | (60.7)                     | (9.3)                      | 21,502             | 1,007                 | 18,880                  | 9,639                   | 4,074                          | 5                        | 1.4                            | 0.4                                  |
| AOL TIME WARNER INC            | Media                        | 15.22              | (1.11)                           | (22.15)                          | na    | na        | (18.6)                     | 7.9                        | 40,961             | (44,574)              | 115,450                 | 65,841                  | 23,275                         | na                       | 1.3                            | 1.6                                  |
| DUKE ENERGY CORP               | Multi-Utilities              | 19.38              | 2.45                             | 1.22                             | -30%  | 16.6      | (35.8)                     | (4.0)                      | 15,663             | 1,034                 | 60,966                  | 17,464                  | 6,482                          | 7                        | 1.1                            | 1.1                                  |
| EXXON MOBIL CORP               | Oil & Gas                    | 36.40              | 2.23                             | 1.69                             | -24%  | 15.5      | (6.4)                      | 3.0                        | 178,909            | 11,011                | 152,644                 | 243,130                 | 11,513                         | 6                        | 3.1                            | 1.4                                  |
| JOHNSON & JOHNSON              | Pharmaceuticals              | 54.35              | 1.87                             | 2.20                             | 18%   | 23.8      | (10.0)                     | 11.0                       | 36,298             | 6,597                 | 40,556                  | 161,355                 | 7,704                          | 18                       | 6.7                            | 4.4                                  |
| MERCK & CO                     | Pharmaceuticals              | 55.58              | 3.18                             | 3.17                             | 0%    | 17.3      | 0.1                        | 0.9                        | 51,790             | 7,150                 | 47,561                  | 124,629                 | 6,559                          | 14                       | 6.6                            | 2.4                                  |
| PFIZER INC                     | Pharmaceuticals              | 31.02              | 1.25                             | 1.48                             | 18%   | 20.4      | (8.7)                      | (1.3)                      | 32,373             | 9,181                 | 46,356                  | 244,886                 | 22,679                         | 28                       | 9.9                            | 7.6                                  |
| ELECTRONIC ARTS INC            | Software                     | 68.55              | (0.08)                           | 0.71                             | na    | 29.3      | 7.1                        | 25.8                       | 2,482              | 317                   | 2,360                   | 9,810                   | 4,084                          | 13                       | 5.7                            | 4.0                                  |
| INTUIT INC                     | Software                     | 45.92              | (0.40)                           | 0.66                             | na    | 41.4      | 5.0                        | 23.8                       | 1,338              | 70                    | 2,963                   | 9,446                   | 4,323                          | 5                        | 4.2                            | 7.0                                  |
| MICROSOFT CORP                 | Software                     | 24.61              | 0.69                             | 0.73                             | 5%    | 27.7      | (3.0)                      | 3.1                        | 28,365             | 7,829                 | 67,646                  | 263,361                 | 62,502                         | 28                       | 4.5                            | 9.3                                  |
| RED HAT INC                    | Software                     | 7.45               | (0.53)                           | (0.83)                           | na    | na        | 54.9                       | n/a                        | 79                 | (120)                 | 370                     | 1,270                   | 1,146                          | na                       | 3.8                            | 16.1                                 |
| SIEBEL SYSTEMS INC             | Software                     | 9.41               | 0.56                             | (0.08)                           | na    | na        | (48.4)                     | 10.6                       | 1,635              | (36)                  | 3,033                   | 4,607                   | 11,146                         | na                       | 2.3                            | 2.8                                  |
| HOME DEPOT INC                 | Specialty Retail             | 32.49              | 1.11                             | 1.30                             | 17%   | 20.2      | (21.4)                     | 4.8                        | 58,247             | 3,664                 | 30,011                  | 74,525                  | 11,954                         | 6                        | 3.6                            | 1.3                                  |
| ALTRIA GROUP INC               | Tobacco                      | 41.30              | 3.92                             | 5.26                             | 34%   | 7.9       | (22.9)                     | 7.9                        | 62,182             | 11,102                | 87,540                  | 83,626                  | 13,192                         | 18                       | 4.1                            | 1.3                                  |
| AT&T WIRELESS SERVICES INC     | Wireless Telecom Svcs        | 7.77               | (0.38)                           | (0.87)                           | na    | na        | (4.2)                      | na                         | 15,631             | (2,205)               | 45,806                  | 21,071                  | 12,640                         | na                       | 0.9                            | 1.3                                  |
| Averages (where appropriate)   |                              |                    |                                  |                                  |       |           |                            |                            |                    |                       |                         |                         |                                |                          |                                |                                      |
| Minimum                        |                              |                    |                                  |                                  | -53%  | (9.5)     | (60.7)                     | (19.6)                     | 6                  | (44,574)              | 152                     | 163                     | 116                            | 16.6                     | (10.9)                         | 0.2                                  |
| Maximum                        |                              |                    |                                  |                                  | 964%  | 124.3     | 105.0                      | 37.3                       | 244,524            | 15,133                | 1,097,190               | 286,838                 | 63,625                         | 16.6                     | 4.3                            | 27.7                                 |
| Standard Deviation             |                              |                    |                                  |                                  | 211%  | 28.6      | 36.8                       | 12.1                       | 50,338             | 8,650                 | 209,018                 | 86,151                  | 17,269                         | 4.3                      | 6.6                            | 6.6                                  |

Source: Standard &amp; Poor's.

## Exhibit 10 Three-Stage Financing of ME2

|                                                            | Year<br>0 | Year<br>1  | Year<br>2  | Year<br>3  | Year<br>4  | Year<br>5  |            |           |
|------------------------------------------------------------|-----------|------------|------------|------------|------------|------------|------------|-----------|
| 1. Revenue Growth Rate                                     |           | 150.0%     | 100.0%     | 100.0%     | 60.00%     | 60.00%     |            |           |
| Revenues                                                   | 500,000   | 1,250,000  | 2,500,000  | 5,000,000  | 8,000,000  | 12,800,000 |            |           |
| Net Income Percent                                         | -50.0%    | -5.0%      | 10.0%      | 15.0%      | 17.0%      | 19.5%      |            |           |
| Net Income                                                 | (250,000) | (62,500)   | 250,000    | 750,000    | 1,360,000  | 2,500,000  |            |           |
| Interest (A/T) @ 7.9%                                      | 0         | 0          | 0          | 0          | 0          | 0          |            |           |
| EBIAT                                                      | (250,000) | (62,500)   | 250,000    | 750,000    | 1,360,000  | 2,500,000  |            |           |
| 2. Working Capital @ 20.0%                                 | 100,000   | 250,000    | 500,000    | 1,000,000  | 1,600,000  | 2,560,000  |            |           |
| Fixed Assets @ 40.0%                                       | 200,000   | 500,000    | 1,000,000  | 2,000,000  | 3,200,000  | 5,120,000  |            |           |
| Free Cash Flow                                             | (549,999) | (512,500)  | (500,000)  | (750,000)  | (440,000)  | (380,000)  |            |           |
| Cum. External Finc. Need                                   | 549,999   | 1,062,499  | 1,562,499  | 2,312,499  | 2,752,499  | 3,132,499  |            |           |
| 3. Debt Issues                                             | 0         | 0          | 0          | 0          | 0          | 0          |            |           |
| Equity Issues                                              | 1,500,000 | 0          | 1,000,000  | 0          | 1,000,000  | 0          |            |           |
| Debt Outstanding                                           | 0         | 0          | 0          | 0          | 0          | 0          |            |           |
| Equity Outstanding                                         | 1,500,000 | 1,500,000  | 2,500,000  | 2,500,000  | 3,500,000  | 3,500,000  |            |           |
| 4. Cash Balance                                            | 950,001   | 437,501    | 937,501    | 187,501    | 747,501    | 367,501    |            |           |
| 5. Final % Ownership Allocated for Management:             |           |            | 0.0%       |            |            |            |            |           |
| 6. Assume: Long-term IRR Required Each Round by Investors  |           |            |            |            |            |            |            |           |
|                                                            | 50.0%     | 45.0%      | 40.0%      | 30.0%      | 25.0%      | 20.0%      |            |           |
| 7. Implied IRR in Each Round, Looking Forward ____ Years:  |           |            |            |            |            |            |            |           |
| 1                                                          | 71.8%     | 61.1%      | 62.4%      | 35.2%      | 25.0%      |            |            |           |
| 2                                                          | 66.4      | 61.7       | 48.2       | 30.0       | 25.0       |            |            |           |
| 3                                                          | 65.0      | 52.4       | 40.0       | 0          | 0          |            |            |           |
| 4                                                          | 57.0      | 45.0       |            |            |            |            |            |           |
| 5                                                          | 50.0      |            |            |            |            |            |            |           |
| 8. Final % Ownership Required by Investors at Terminal PER |           |            |            |            |            |            | Management | Total     |
| 5                                                          | 91.1%     | 0.0%       | 22.0%      | 0.0%       | 10.0%      | 0.0%       | 0.0%       | 123.1%    |
| 10                                                         | 45.6      | 0.0        | 11.0       | 0.0        | 5.0        | 0.0        | 0.0        | 61.5      |
| 15                                                         | 30.4      | 0.0        | 7.3        | 0.0        | 3.3        | 0.0        | 0.0        | 41.0      |
| 20                                                         | 22.8      | 0.0        | 5.5        | 0.0        | 2.5        | 0.0        | 0.0        | 30.8      |
| 9. Eventual Retention % Faced by an Investor in Each Year: |           |            |            |            |            |            |            |           |
| 5                                                          | 68.0%     | 68.0%      | 90.0%      | 90.0%      | 100.0%     | 100.0%     |            |           |
| 10                                                         | 84.0      | 84.0       | 95.0       | 95.0       | 100.0      | 100.0      |            |           |
| 15                                                         | 89.3      | 89.3       | 96.7       | 96.7       | 100.0      | 100.0      |            |           |
| 20                                                         | 92.0      | 92.0       | 97.5       | 97.5       | 100.0      | 100.0      |            |           |
| 10. Current (at financing) % Ownership Required            |           |            |            |            |            |            |            |           |
| 5                                                          | 133.9%    | 0.0%       | 24.4%      | 0.0%       | 10.0%      | 0.0%       |            |           |
| 10                                                         | 54.2      | 0.0        | 11.6       | 0.0        | 5.0        | 0.0        |            |           |
| 15                                                         | 34.0      | 0.0        | 7.6        | 0.0        | 3.3        | 0.0        |            |           |
| 20                                                         | 24.8      | 0.0        | 5.6        | 0.0        | 2.5        | 0.0        |            |           |
| 11. Shares to Issue, If Pre-Financing Shares = 1,000,000   |           |            |            |            |            |            | Management | Total     |
| 5                                                          | NA        | NA         | NA         | NA         | NA         | NA         | NA         | NA        |
| 10                                                         | 1,184,626 | 0          | 285,376    | 0          | 130,000    | 0          | 0          | 1,600,000 |
| 15                                                         | 515,055   | 0          | 124,077    | 0          | 56,522     | 0          | 0          | 695,653   |
| 20                                                         | 329,063   | 0          | 79,271     | 0          | 36,111     | 0          | 0          | 444,445   |
| 12. Cumulative Shares Outstanding                          |           |            |            |            |            |            |            |           |
| 5                                                          | NA        | NA         | NA         | NA         | NA         | NA         |            |           |
| 10                                                         | 2,184,626 | 2,184,626  | 2,470,002  | 2,470,002  | 2,600,003  | 2,600,003  |            |           |
| 15                                                         | 1,515,055 | 1,515,055  | 1,639,131  | 1,639,131  | 1,695,653  | 1,695,653  |            |           |
| 20                                                         | 1,329,063 | 1,329,063  | 1,408,334  | 1,408,334  | 1,444,445  | 1,444,445  |            |           |
| 13. Price Per Share                                        |           |            |            |            |            |            |            |           |
| 5                                                          | NA        | NA         | NA         | NA         | NA         | NA         |            |           |
| 10                                                         | \$1.27    | NA         | \$3.50     | NA         | \$ 7.69    | \$ 9.62    |            |           |
| 15                                                         | 2.91      | NA         | 8.06       | NA         | 17.69      | 22.12      |            |           |
| 20                                                         | 4.56      | NA         | 12.61      | NA         | 27.69      | 34.62      |            |           |
| 14. Implied Valuation (Post-Money)                         |           |            |            |            |            |            |            |           |
| 5                                                          | 1,120,132 | NA         | 4,099,854  | NA         | 10,000,000 | 12,500,000 |            |           |
| 10                                                         | 2,766,222 | 4,751,944  | 8,655,248  | 14,053,254 | 20,000,000 | 25,000,000 |            |           |
| 15                                                         | 4,412,313 | 7,579,675  | 13,210,641 | 21,449,704 | 30,000,000 | 37,500,000 |            |           |
| 20                                                         | 6,058,403 | 10,407,405 | 17,766,035 | 28,846,154 | 40,000,000 | 50,000,000 |            |           |

## Exhibit 11 Single-Stage Financing of ME2

|                                                            | Year<br>0 | Year<br>1  | Year<br>2  | Year<br>3  | Year<br>4  | Year<br>5  |            |           |
|------------------------------------------------------------|-----------|------------|------------|------------|------------|------------|------------|-----------|
| 1. Revenue Growth Rate                                     |           | 150.0%     | 100.0%     | 100.0%     | 60.00%     | 60.00%     |            |           |
| Revenues                                                   | 500,000   | 1,250,000  | 2,500,000  | 5,000,000  | 8,000,000  | 12,800,000 |            |           |
| Net Income Percent                                         | -50.0%    | -5.0%      | 10.0%      | 15.0%      | 17.0%      | 19.5%      |            |           |
| Net Income                                                 | (250,000) | (62,500)   | 250,000    | 750,000    | 1,360,000  | 2,500,000  |            |           |
| Interest (A/T) @ 7.9%                                      | 0         | 0          | 0          | 0          | 0          | 0          |            |           |
| EBIAT                                                      | (250,000) | (62,500)   | 250,000    | 750,000    | 1,360,000  | 2,500,000  |            |           |
| 2. Working Capital @ 20.0%                                 | 100,000   | 250,000    | 500,000    | 1,000,000  | 1,600,000  | 2,560,000  |            |           |
| Fixed Assets @ 40.0%                                       | 200,000   | 500,000    | 1,000,000  | 2,000,000  | 3,200,000  | 5,120,000  |            |           |
| Free Cash Flow                                             | (549,999) | (512,500)  | (500,000)  | (750,000)  | (440,000)  | (380,000)  |            |           |
| Cum. External Finc. Need                                   | 549,999   | 1,062,499  | 1,562,499  | 2,312,499  | 2,752,499  | 3,132,499  |            |           |
| 3. Debt Issues                                             | 0         | 0          | 0          | 0          | 0          | 0          |            |           |
| Equity Issues                                              | 3,500,000 | 0          | 1,000,000  | 0          | 1,000,000  | 0          |            |           |
| Debt Outstanding                                           | 0         | 0          | 0          | 0          | 0          | 0          |            |           |
| Equity Outstanding                                         | 3,500,000 | 1,500,000  | 2,500,000  | 2,500,000  | 3,500,000  | 3,500,000  |            |           |
| 4. Cash Balance                                            | 2,950,001 | 2,437,501  | 1,937,501  | 1,187,501  | 747,501    | 367,501    |            |           |
| 5. Final % Ownership Allocated for Management:             |           |            | 0.0%       |            |            |            |            |           |
| 6. Assume: Long-term IRR Required Each Round by Investors  |           |            |            |            |            |            |            |           |
|                                                            | 50.0%     | 45.0%      | 40.0%      | 30.0%      | 25.0%      | 20.0%      |            |           |
| 7. Implied IRR in Each Round, Looking Forward ____ Years:  |           |            |            |            |            |            |            |           |
| 1                                                          | 71.8%     | 61.1%      | 62.4%      | 35.2%      | 25.0%      |            |            |           |
| 2                                                          | 66.4      | 61.7       | 48.2       | 30.0       |            |            |            |           |
| 3                                                          | 65.0      | 52.4       | 40.0       |            |            |            |            |           |
| 4                                                          | 57.0      | 45.0       |            |            |            |            |            |           |
| 5                                                          | 50.0      |            |            |            |            |            |            |           |
| 8. Final % Ownership Required by Investors at Terminal PER |           |            |            |            |            |            | Management | Total     |
| 5                                                          | 212.6%    | 0.0%       | 0.0%       | 0.0%       | 0.0%       | 0.0%       | 0.0%       | 212.6%    |
| 10                                                         | 106.3     | 0.0        | 0.0        | 0.0        | 0.0        | 0.0        | 0.0        | 106.3     |
| 15                                                         | 70.9      | 0.0        | 0.0        | 0.0        | 0.0        | 0.0        | 0.0        | 70.9      |
| 20                                                         | 53.2      | 0.0        | 0.0        | 0.0        | 0.0        | 0.0        | 0.0        | 53.2      |
| 9. Eventual Retention % Faced by an Investor in Each Year: |           |            |            |            |            |            |            |           |
| 5                                                          | 100.0%    | 100.0%     | 100.0%     | 100.0%     | 100.0%     | 100.0%     |            |           |
| 10                                                         | 100.0     | 100.0      | 100.0      | 100.0      | 100.0      | 100.0      |            |           |
| 15                                                         | 100.0     | 100.0      | 100.0      | 100.0      | 100.0      | 100.0      |            |           |
| 20                                                         | 100.0     | 100.0      | 100.0      | 100.0      | 100.0      | 100.0      |            |           |
| 10. Current (at financing) % Ownership Required            |           |            |            |            |            |            |            |           |
| 5                                                          | 212.6%    | 0.0%       | 0.0%       | 0.0%       | 0.0%       | 0.0%       |            |           |
| 10                                                         | 106.2     | 0.0        | 0.0        | 0.0        | 0.0        | 0.0        |            |           |
| 15                                                         | 70.9      | 0.0        | 0.0        | 0.0        | 0.0        | 0.0        |            |           |
| 20                                                         | 53.2      | 0.0        | 0.0        | 0.0        | 0.0        | 0.0        |            |           |
| 11. Shares to Issue, If Pre-Financing Shares = 1,000,000   |           |            |            |            |            |            | Management | Total     |
| 5                                                          | NA        | NA         | NA         | NA         | NA         | NA         | NA         | NA        |
| 10                                                         | NA        | 0          | 0          | 0          | 0          | 0          | 0          | NA        |
| 15                                                         | 2,433,476 | 0          | 0          | 0          | 0          | 0          | 0          | 2,433,476 |
| 20                                                         | 1,134,757 | 0          | 0          | 0          | 0          | 0          | 0          | 1,134,757 |
| 12. Cumulative Shares Outstanding                          |           |            |            |            |            |            |            |           |
| 5                                                          | NA        | NA         | NA         | NA         | NA         | NA         |            |           |
| 10                                                         | NA        | NA         | NA         | NA         | NA         | NA         |            |           |
| 15                                                         | 3,433,476 | 3,433,476  | 3,433,476  | 3,433,476  | 3,433,476  | 3,433,476  |            |           |
| 20                                                         | 2,134,757 | 2,134,757  | 2,134,757  | 2,134,757  | 2,134,757  | 2,134,757  |            |           |
| 13. Price Per Share                                        |           |            |            |            |            |            |            |           |
| 5                                                          | NA        | NA         | NA         | NA         | NA         | NA         |            |           |
| 10                                                         | NA        | NA         | NA         | NA         | NA         | NA         |            |           |
| 15                                                         | \$1.44    | NA         | NA         | NA         | NA         | NA         | \$10.92    |           |
| 20                                                         | 3.08      | NA         | NA         | NA         | NA         | NA         | 23.42      |           |
| 14. Implied Valuation (Post-Money)                         |           |            |            |            |            |            |            |           |
| 5                                                          | 1,646,091 | NA         | NA         | NA         | NA         | 12,500,000 |            |           |
| 10                                                         | 3,292,181 | NA         | NA         | NA         | NA         | 25,000,000 |            |           |
| 15                                                         | 4,938,272 | 8,483,191  | 13,666,181 | 22,189,349 | 30,000,000 | 37,500,000 |            |           |
| 20                                                         | 6,584,362 | 11,310,922 | 18,221,574 | 19,585,799 | 40,000,000 | 50,000,000 |            |           |

**Exhibit 12** Three-Stage Financing of ME2, With Management Equity

|                                                            | Year<br>0 | Year<br>1 | Year<br>2  | Year<br>3  | Year<br>4  | Year<br>5  |            |        |
|------------------------------------------------------------|-----------|-----------|------------|------------|------------|------------|------------|--------|
| 1. Revenue Growth Rate                                     |           | 150.0%    | 100.0%     | 100.0%     | 60.00%     | 60.00%     |            |        |
| Revenues                                                   | 500,000   | 1,250,000 | 2,500,000  | 5,000,000  | 8,000,000  | 12,800,000 |            |        |
| Net Income Percent                                         | -50.0%    | -5.0%     | 10.0%      | 15.0%      | 17.0%      | 19.5%      |            |        |
| Net Income                                                 | (250,000) | (62,500)  | 250,000    | 750,000    | 1,360,000  | 2,500,000  |            |        |
| Interest (A/T) @ 7.9%                                      | 0         | 0         | 0          | 0          | 0          | 0          |            |        |
| EBIAT                                                      | (250,000) | (62,500)  | 250,000    | 750,000    | 1,360,000  | 2,500,000  |            |        |
| 2. Working Capital @ 20.0%                                 | 100,000   | 250,000   | 500,000    | 1,000,000  | 1,600,000  | 2,560,000  |            |        |
| Fixed Assets @ 40.0%                                       | 200,000   | 500,000   | 1,000,000  | 2,000,000  | 3,200,000  | 5,120,000  |            |        |
| Free Cash Flow                                             | (549,999) | (512,500) | (500,000)  | (750,000)  | (440,000)  | (380,000)  |            |        |
| Cum. External Finc. Need                                   | 549,999   | 1,062,499 | 1,562,499  | 2,312,499  | 2,752,499  | 3,132,499  |            |        |
| 3. Debt Issues                                             | 0         | 0         | 0          | 0          | 0          | 0          |            |        |
| Equity Issues                                              | 1,500,000 | 0         | 1,000,000  | 0          | 1,000,000  | 0          |            |        |
| Debt Outstanding                                           | 0         | 0         | 0          | 0          | 0          | 0          |            |        |
| Equity Outstanding                                         | 1,500,000 | 1,500,000 | 2,500,000  | 2,500,000  | 3,500,000  | 3,500,000  |            |        |
| 4. Cash Balance                                            | 950,001   | 437,501   | 937,501    | 187,501    | 747,501    | 367,501    |            |        |
| 5. Final % Ownership Allocated for Management:             |           |           | 15.0%      |            |            |            |            |        |
| 6. Assume: Long-term IRR Required Each Round by Investors  |           |           |            |            |            |            |            |        |
|                                                            | 50.0%     | 45.0%     | 40.0%      | 30.0%      | 25.0%      | 20.0%      |            |        |
| 7. Implied IRR in Each Round, Looking Forward ____ Years:  |           |           |            |            |            |            |            |        |
| 1                                                          | 71.8%     | 61.1%     | 62.4%      | 35.2%      | 20.0%      |            |            |        |
| 2                                                          | 66.4      | 61.7      | 48.2       | 30.0       |            |            |            |        |
| 3                                                          | 65.0      | 52.4      | 40.0       |            |            |            |            |        |
| 4                                                          | 57.0      | 45.0      |            |            |            |            |            |        |
| 5                                                          | 50.0      |           |            |            |            |            |            |        |
| 8. Final % Ownership Required by Investors at Terminal PER |           |           |            |            |            |            | Management | Total  |
| 5                                                          | 91.1%     | 0.0%      | 22.0%      | 0.0%       | 10.0%      | 15.0%      | 15.0%      | 138.1% |
| 10                                                         | 45.6      | 0.0       | 11.0       | 0.0        | 5.0        | 15.0       | 15.0       | 76.5   |
| 15                                                         | 30.4      | 0.0       | 7.3        | 0.0        | 3.3        | 15.0       | 15.0       | 56.0   |
| 20                                                         | 22.8      | 0.0       | 5.5        | 0.0        | 2.5        | 15.0       | 15.0       | 45.8   |
| 9. Eventual Retention % Faced by an Investor in Each Year: |           |           |            |            |            |            |            |        |
| 5                                                          | 53.0%     | 53.0%     | 75.0%      | 75.0%      | 85.0%      | 100.0%     |            |        |
| 10                                                         | 69.0      | 69.0      | 80.0       | 80.0       | 85.0       | 100.0      |            |        |
| 15                                                         | 74.3      | 74.3      | 81.7       | 81.7       | 85.0       | 100.0      |            |        |
| 20                                                         | 77.0      | 77.0      | 82.5       | 82.5       | 85.0       | 100.0      |            |        |
| 10. Current (at financing) % Ownership Required            |           |           |            |            |            |            |            |        |
| 5                                                          | 171.8%    | 0.0%      | 29.0%      | 0.0%       | 11.8%      | 15.0%      |            |        |
| 10                                                         | 66.0      | 0.0       | 13.0       | 0.0        | 5.9        | 15.0       |            |        |
| 15                                                         | 40.9      | 0.0       | 9.0        | 0.0        | 3.9        | 15.0       |            |        |
| 20                                                         | 29.6      | 0.0       | 6.7        | 0.0        | 2.9        | 15.0       |            |        |
| 11. Shares to Issue, If Pre-Financing Shares = 1,000,000   |           |           |            |            |            |            | Management | Total  |
| 5                                                          | NA        | NA        | NA         | NA         | NA         | NA         | NA         | NA     |
| 10                                                         | 1,942,011 | NA        | 467,830    | 0          | 213,115    | 639,345    | 3,262,302  |        |
| 15                                                         | 690,744   | 0         | 166,400    | 0          | 75,802     | 341,108    | 1,274,054  |        |
| 20                                                         | 420,080   | 0         | 101,197    | 0          | 46,099     | 276,596    | 843,972    |        |
| 12. Cumulative Shares Outstanding                          |           |           |            |            |            |            |            |        |
| 5                                                          | NA        | NA        | NA         | NA         | NA         | NA         |            |        |
| 10                                                         | 2,942,011 | 2,942,011 | 3,409,842  | 3,409,842  | 3,622,957  | 4,262,302  |            |        |
| 15                                                         | 1,690,744 | 1,690,744 | 1,857,144  | 1,857,144  | 1,932,946  | 1,274,054  |            |        |
| 20                                                         | 1,420,080 | 1,420,080 | 1,521,277  | 1,521,277  | 1,567,376  | 1,843,972  |            |        |
| 13. Price Per Share                                        |           |           |            |            |            |            |            |        |
| 5                                                          | NA        | NA        | NA         | NA         | NA         | NA         |            |        |
| 10                                                         | \$0.77    | NA        | \$2.14     | NA         | \$4.69     | \$5.87     |            |        |
| 15                                                         | 2.17      | NA        | 6.01       | NA         | 13.19      | \$16.49    |            |        |
| 20                                                         | 3.57      | NA        | 9.88       | NA         | 21.69      | 27.12      |            |        |
| 14. Implied Valuation (Post-Money)                         |           |           |            |            |            |            |            |        |
| 5                                                          | 873,218   | NA        | 3,416,545  | NA         | 8,500,000  | 12,500,000 |            |        |
| 10                                                         | 2,272,395 | 3,903,625 | 7,288,630  | 11,834,320 | 17,000,000 | 25,000,000 |            |        |
| 15                                                         | 3,671,572 | 6,307,196 | 11,160,714 | 18,121,302 | 25,000,000 | 37,500,000 |            |        |
| 20                                                         | 5,070,749 | 8,710,767 | 15,032,799 | 24,408,284 | 34,000,000 | 50,000,000 |            |        |

## Exhibit 13 Final Ownership Shares in Selected Initial Public Offerings

| Company                                        | Investor  | Date    | Amount Raised (\$000) | Cum. Funding (\$000) | Stock Rec'd (000) | Total Shares Out'g (000) | % Owner'p Acquired | Fully Diluted Valuation (\$000) | Price Per Share (\$) | Ending Owner' Ship |
|------------------------------------------------|-----------|---------|-----------------------|----------------------|-------------------|--------------------------|--------------------|---------------------------------|----------------------|--------------------|
| Apple Computer<br>(Personal Computers)         | Founders  | Mar-77  | 1                     | 1                    | 16,640            | 16,640                   | 100.0%             | 1                               | 0.00                 | 30.7%              |
|                                                | Founders  | Nov-77  | 115                   | 115                  | 10,480            | 27,120                   | 38.6%              | 298                             | 0.01                 | 19.3%              |
|                                                | Founders  | Jul-78  | 426                   | 541                  | 4,736             | 31,856                   | 14.9%              | 2,867                           | 0.09                 | 8.7%               |
|                                                | Venture 1 | Jan-78  | 518                   | 1,059                | 5,520             | 37,376                   | 14.8%              | 3,364                           | 0.09                 | 10.2%              |
|                                                | Venture 2 | Sep-78  | 704                   | 1,763                | 2,503             | 39,879                   | 6.3%               | 13,406                          | 0.28                 | 4.6%               |
|                                                | Venture 3 | Aug-79  | 2,331                 | 4,298                | 2,400             | 43,306                   | 5.5%               | 59,834                          | 0.97                 | 4.4%               |
|                                                | IPO       | Dec-80  | 88,000                | 94,487               | 4,000             | 54,215                   | 7.4%               | 1,412,370                       | 22.00                | 7.4%               |
| Cray Research<br>(Computers)                   | Founders  | Aug-72  | 2,550                 | 2,550                | 2,869             | 2,869                    | 100.0%             | 2,794                           | 0.89                 | 21.1%              |
|                                                | Venture 1 | Jan-74  | 2,675                 | 5,225                | 2,006             | 4,875                    | 41.1%              | 6,843                           | 1.33                 | 14.8%              |
|                                                | Venture 2 | Jan-75  | 642                   | 5,867                | 387               | 5,302                    | 7.3%               | 9,557                           | 1.66                 | 2.9%               |
|                                                | Venture 3 | Apr-75  | 2,720                 | 8,587                | 1,530             | 6,832                    | 22.4%              | 12,971                          | 1.78                 | 11.3%              |
|                                                | IPO       | Mar-76  | 10,890                | 19,512               | 4,950             | 11,783                   | 42.0%              | 26,924                          | 2.20                 | 36.5%              |
|                                                | SPO       | Oct-80  | 38,000                | 57,341               | 1,093             | 13,567                   | 8.1%               | 446,104                         | 31.67                | 8.1%               |
| Genentech<br>(Biotechnology)                   | Founders  | Jan-76  | 126                   | 126                  | 3,200             | 3,200                    | 100.0%             | 126                             | 0.04                 | 41.4%              |
|                                                | Venture 1 | Apr-76  | 850                   | 976                  | 1,180             | 4,280                    | 27.6%              | 3,154                           | 0.72                 | 15.3%              |
|                                                | Venture 2 | May-78  | 950                   | 1,923                | 475               | 4,945                    | 9.6%               | 9,890                           | 2.00                 | 6.1%               |
|                                                | Corporate | Sep-79  | 10,000                | 11,953               | 1,000             | 6,348                    | 15.8%              | 63,480                          | 10.00                | 12.9%              |
|                                                | IPO       | Oct-80  | 38,500                | 50,607               | 1,100             | 7,724                    | 14.2%              | 298,340                         | 35.00                | 14.2%              |
| JetBlue<br>(Airline)                           | Seed      | Oct-98  | 4,900                 | 4,900                | na                | na                       | 100.0%             | na                              | na                   | 1.8%               |
|                                                | Venture 1 | Dec-98  | 129,820               | 134,720              | na                | na                       | 85.2%              | 152,320                         | na                   | 59.8%              |
|                                                | Venture 2 | Aug-00  | 44,910                | 179,620              | na                | na                       | 17.4%              | 258,790                         | na                   | 15.1%              |
|                                                | IPO       | Apr-02  | 158,400               | 338,020              | na                | na                       | 13.0%              | 1,222,390                       | na                   | 13.0%              |
| Seagate<br>(Computer Periph'ls)                | Founders  | Oct-79  | 161                   | 161                  | 11,723            | 11,723                   | 100.0%             | 161                             | 0.01                 | 64.1%              |
|                                                | Venture 1 | Jun-80  | 1,000                 | 1,161                | 3,125             | 14,848                   | 21.0%              | 4,949                           | 0.32                 | 17.1%              |
|                                                | IPO       | Sep-81  | 25,000                | 26,161               | 2,500             | 18,277                   | 13.7%              | 186,620                         | 10.00                | 13.7%              |
| Lotus Development<br>(Software)                | Founders  | Apr-82  | 13                    | 13                   | 4,410             | 4,410                    | 100.0%             | 13                              | 0.003                | 27.9%              |
|                                                | Venture 1 | Apr-82  | 1,000                 | 1,013                | 3,500             | 7,910                    | 43.5%              | 2,296                           | 0.29                 | 22.1%              |
|                                                | Venture 2 | Dec-82  | 3,705                 | 4,718                | 3,592             | 11,502                   | 28.0%              | 13,232                          | 1.03                 | 22.7%              |
|                                                | IPO       | Oct-83  | 53,820                | 58,538               | 2,990             | 14,492                   | 18.9%              | 287,742                         | 18.00                | 18.9%              |
| Avid Technology, Inc.<br>(Multimedia Software) | Founders  | Jan-86  | 1                     | 1                    | 600               | 600                      | 100.0%             | 1                               | 0.002                | 13.1%              |
|                                                | Venture 1 | Feb-86  | 500                   | 501                  | 500               | 1,100                    | 37.0%              | 1,350                           | 1.00                 | 10.9%              |
|                                                | Venture 2 | Feb -87 | 2,050                 | 2,551                | 854               | 1,954                    | 37.9%              | 5,410                           | 2.40                 | 18.7%              |
|                                                | Venture 3 | Feb -88 | 3,000                 | 5,551                | 750               | 2,704                    | 24.6%              | 12,217                          | 4.00                 | 16.5%              |
|                                                | Venture 4 | Feb -89 | 20,000                | 25,551               | 500               | 3,204                    | 13.9%              | 144,167                         | 40.00                | 11.0%              |
|                                                | IPO       | Mar-93  | 56,000                | 81,551               | 2,800             | 9,425                    | 29.7%              | 188,500                         | 20.00                | 29.7%              |

Notes: All figures (000) except per share amounts.

Venture 1, etc., represent round of venture capital financing.

IPO = Initial Public Offering

SPC = Secondary Public Offering

LTM = Last Twelve Months (e.g., sales or earnings per share—EPS)

Source: Prospectuses, Annual Reports, casewriter estimates, VentureXpert, VentureSource.

## Exhibit 14 ME2 Financing, Sensitivity Model

|                                                                                                                                                                                                  |                                                                                                             | Year<br>0 | Year<br>1 | Year<br>2  | Year<br>3  | Year<br>4  | Year<br>5  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-----------|-----------|------------|------------|------------|------------|
| 1.                                                                                                                                                                                               | Final % Ownership Required by Investors if Term. PER = 15 and if IRR =                                      |           |           |            |            |            |            |
|                                                                                                                                                                                                  | 20%                                                                                                         | 10.0%     | 0.0%      | 4.6%       | 0.0%       | 3.2%       | 0.0%       |
|                                                                                                                                                                                                  | 25                                                                                                          | 12.2      | 0.0       | 5.2        | 0.0        | 3.3        | 0.0        |
|                                                                                                                                                                                                  | 30                                                                                                          | 14.9      | 0.0       | 5.9        | 0.0        | 3.5        | 0.0        |
|                                                                                                                                                                                                  | 40                                                                                                          | 21.5      | 0.0       | 7.3        | 0.0        | 3.7        | 0.0        |
|                                                                                                                                                                                                  | 50                                                                                                          | 30.4      | 0.0       | 9.0        | 0.0        | 4.0        | 0.0        |
|                                                                                                                                                                                                  | 60                                                                                                          | 41.9      | 0.0       | 10.9       | 0.0        | 4.3        | 0.0        |
| 2.                                                                                                                                                                                               | Current % Ownership Required at <i>This</i> Round IRR = ____, and Terminal PER =                            |           |           |            |            |            |            |
|                                                                                                                                                                                                  | 20.0%                                                                                                       | 5         | 10        | 15         | 20         | 25         | 30         |
|                                                                                                                                                                                                  | 25.0                                                                                                        | 56.3%     | 21.6%     | 13.4%      | 9.7%       | 7.6%       | 6.2%       |
|                                                                                                                                                                                                  | 30.0                                                                                                        | 69.0      | 26.5      | 16.4       | 11.9       | 9.3        | 7.7        |
|                                                                                                                                                                                                  | 40.0                                                                                                        | 84.0      | 32.3      | 20.0       | 14.5       | 11.3       | 9.3        |
|                                                                                                                                                                                                  | 50.0                                                                                                        | 121.7     | 46.8      | 28.9       | 21.0       | 16.4       | 13.5       |
|                                                                                                                                                                                                  | 60.0                                                                                                        | 171.8     | 66.0      | 40.9       | 29.6       | 23.2       | 19.1       |
|                                                                                                                                                                                                  |                                                                                                             | 237.2     | 91.1      | 56.4       | 40.8       | 32.0       | 26.3       |
| 3.                                                                                                                                                                                               | Implied Valuation                                                                                           |           |           |            |            |            |            |
|                                                                                                                                                                                                  | 5                                                                                                           | 10        | 15        | 20         | 25         | 30         |            |
|                                                                                                                                                                                                  | 20.0%                                                                                                       | 2,664,850 | 6,934,799 | 11,204,749 | 15,474,698 | 19,744,647 | 24,014,596 |
|                                                                                                                                                                                                  | 25.0                                                                                                        | 2,172,846 | 5,654,446 | 9,136,046  | 12,617,646 | 16,099,246 | 19,580,846 |
|                                                                                                                                                                                                  | 30.0                                                                                                        | 1,785,921 | 4,647,543 | 7,509,164  | 10,370,785 | 13,232,407 | 16,094,028 |
|                                                                                                                                                                                                  | 40.0                                                                                                        | 1,232,931 | 3,208,485 | 5,184,038  | 7,159,591  | 9,135,145  | 11,110,698 |
|                                                                                                                                                                                                  | 50.0                                                                                                        | 873,218   | 1,272,395 | 3,671,572  | 5,070,749  | 6,469,926  | 7,869,103  |
|                                                                                                                                                                                                  | 60.0                                                                                                        | 632,381   | 1,645,660 | 2,658,939  | 3,672,218  | 4,685,497  | 5,698,776  |
| 4.                                                                                                                                                                                               | Price Corresponding to the Average % Ownerships                                                             |           |           |            |            |            |            |
|                                                                                                                                                                                                  | 5                                                                                                           | 10        | 15        | 20         | 25         | 30         |            |
|                                                                                                                                                                                                  | 20.0%                                                                                                       | \$ 1.16   | \$5.43    | \$9.70     | \$13.97    | \$18.24    | \$22.51    |
|                                                                                                                                                                                                  | 25.0                                                                                                        | 0.67      | 4.15      | 7.64       | 11.12      | 14.60      | 18.08      |
|                                                                                                                                                                                                  | 30.0                                                                                                        | 0.29      | 3.15      | 6.01       | 8.78       | 11.73      | 14.59      |
|                                                                                                                                                                                                  | 40.0                                                                                                        | (0.27)    | 1.71      | 3.68       | 5.66       | 7.64       | 9.61       |
|                                                                                                                                                                                                  | 50.0                                                                                                        | (0.63)    | 0.77      | 2.17       | 3.57       | 4.97       | 6.37       |
|                                                                                                                                                                                                  | 60.0                                                                                                        | (0.87)    | 0.15      | 1.16       | 2.17       | 3.19       | 4.20       |
| 5.                                                                                                                                                                                               | Current % Ownership Required at <i>This</i> Round IRR = (down), and <i>Next</i> Round IRR = (across). PER = |           |           |            |            |            |            |
|                                                                                                                                                                                                  | 20.0%                                                                                                       | 20.0%     | 25.0%     | 30.0%      | 40.0%      | 50.0%      | 60.0%      |
|                                                                                                                                                                                                  | 25.0                                                                                                        | 12.9      | 13.0      | 13.1%      | 13.4       | 13.7       | 14.1       |
|                                                                                                                                                                                                  | This round < 30.0                                                                                           | 15.8      | 16.0      | 16.1       | 16.4       | 16.8       | 17.3       |
|                                                                                                                                                                                                  | 40.0                                                                                                        | 19.3      | 19.4      | 19.6       | 20.0       | 20.4       | 21.0       |
|                                                                                                                                                                                                  | 50.0                                                                                                        | 27.9      | 28.1      | 28.4       | 28.9       | 29.6       | 30.4       |
|                                                                                                                                                                                                  | 60.0                                                                                                        | 39.4      | 39.7      | 40.1       | 40.9       | 41.8       | 42.9       |
|                                                                                                                                                                                                  |                                                                                                             | 54.4      | 54.9      | 55.3       | 56.4       | 57.7       | 59.3       |
| 6.                                                                                                                                                                                               | Price Corresponding to Above % Ownerships                                                                   |           |           |            |            |            |            |
|                                                                                                                                                                                                  | 20.0%                                                                                                       | 20.0%     | 25.0%     | 30.0%      | 40.0%      | 50.0%      | 60.0%      |
|                                                                                                                                                                                                  | 20.0%                                                                                                       | \$10.11   | \$10.02   | \$9.92     | \$9.70     | \$9.45     | \$9.16     |
|                                                                                                                                                                                                  | 25.0                                                                                                        | 7.97      | 7.90      | 7.82       | 7.64       | 7.43       | 7.19       |
|                                                                                                                                                                                                  | This round < 30.0                                                                                           | 6.28      | 6.22      | 6.16       | 6.01       | 5.84       | 5.65       |
|                                                                                                                                                                                                  | 40.0                                                                                                        | 3.87      | 3.83      | 3.79       | 3.68       | 3.57       | 3.43       |
|                                                                                                                                                                                                  | 50.0                                                                                                        | 2.31      | 2.28      | 2.24       | 2.17       | 2.09       | 1.99       |
|                                                                                                                                                                                                  | 60.0                                                                                                        | 1.26      | 1.23      | 1.21       | 1.16       | 1.10       | 1.03       |
| (The following two tables assume that the PER used to price the deal will actually determine terminal value. The IRR listed is the expected return, but does not affect the price of the stock.) |                                                                                                             |           |           |            |            |            |            |
| 7.                                                                                                                                                                                               | IRR for <i>Each</i> round, if Net Income Varies From Plan by Adjustment Factor =                            |           |           |            |            |            |            |
|                                                                                                                                                                                                  | 0.25                                                                                                        | 13.7%     | 2.5%      | -11.8%     | -35.0%     | -68.8%     |            |
|                                                                                                                                                                                                  | 0.50                                                                                                        | 30.6      | 21.9      | 11.1       | -8.1       | -37.5      |            |
|                                                                                                                                                                                                  | 0.75                                                                                                        | 41.6      | 34.9      | 27.2       | 12.6       | -6.3       |            |
|                                                                                                                                                                                                  | 1.00                                                                                                        | 50.0      | 45.0      | 40.0       | 30.0       | 25.0       |            |
|                                                                                                                                                                                                  | 1.50                                                                                                        | 62.7      | 60.5      | 60.3       | 59.2       | 87.5       |            |
|                                                                                                                                                                                                  | 2.00                                                                                                        | 72.3      | 72.4      | 76.4       | 83.8       | 150.0      |            |
|                                                                                                                                                                                                  | 3.00                                                                                                        | 86.9      | 90.8      | 101.9      | 125.2      | 275.0      |            |

## Exhibit 14 ME2 Financing, Sensitivity Model (Continued)

|     |                                                                                                                | Year<br>0   | Year<br>1 | Year<br>2   | Year<br>3  | Year<br>4   | Year<br>5  | Total     |
|-----|----------------------------------------------------------------------------------------------------------------|-------------|-----------|-------------|------------|-------------|------------|-----------|
| 8.  | Required Terminal Net Income to Achieve IRR:                                                                   |             |           |             |            |             |            |           |
|     | 20.0%                                                                                                          | 819,200     | 1,172,716 | 1,574,344   | 2,130,178  | 2,400,000   | 2,500,000  |           |
|     | 25.0                                                                                                           | 1,004,694   | 1,380,728 | 1,779,451   | 2,311,391  | 2,500,000   | 2,500,000  |           |
|     | 30.0                                                                                                           | 1,222,364   | 1,615,256 | 2,001,640   | 2,500,000  | 2,600,000   | 2,500,000  |           |
|     | 40.0                                                                                                           | 1,770,614   | 2,172,602 | 2,500,000   | 2,899,408  | 2,800,000   | 2,500,000  |           |
|     | 50.0                                                                                                           | 2,500,000   | 2,863,077 | 3,074,891   | 3,328,402  | 3,000,000   | 2,500,000  |           |
|     | 60.0                                                                                                           | 3,452,102   | 3,706,363 | 3,731,778   | 3,786,982  | 3,200,000   | 2,500,000  |           |
| 9.  | Assumptions:                                                                                                   |             |           |             |            |             | Management | Total     |
|     | New Shares                                                                                                     | 690,744     | 0         | 166,400     | 0          | 75,802      | 341,810    | 2,274,054 |
|     | Price per Share                                                                                                | \$2.17      | NA        | \$6.01      | NA         | \$13.19     |            |           |
|     | = Equity Raised                                                                                                | \$1,500,000 | NA        | \$1,000,000 | NA         | \$1,000,000 | \$0        |           |
| 10. | Rate of Return for <i>Each</i> Round if Final PER =                                                            |             |           |             |            |             |            |           |
|     | 10                                                                                                             | 38.3%       | NA        | 22.3%       | NA         | -16.7%      |            |           |
|     | 15                                                                                                             | 50.0        | NA        | 40.0        | NA         | 25.0        |            |           |
|     | 20                                                                                                             | 58.9        | NA        | 54.1        | NA         | 66.7        |            |           |
|     | 25                                                                                                             | 66.1        | NA        | 66.0        | NA         | s108.3      |            |           |
| 11. | Rate of Return to <i>This</i> Round, if Net Income Differs from Plan by Factor of:                             |             |           |             |            |             |            |           |
|     |                                                                                                                | 0.25        | 0.50      | 0.75        | 1.00       | 1.50        | 2.00       | 3.00      |
|     | 10                                                                                                             | 4.8%        | 20.4%     | 30.6%       | 38.3%      | 50.0%       | 58.9%      | 72.3%     |
|     | PER < 15                                                                                                       | 13.7        | 30.6      | 41.6        | 50.0       | 62.7        | 72.3       | 86.9      |
|     | 20                                                                                                             | 20.4        | 38.3      | 50.0        | 58.9       | 72.3        | 82.5       | 97.9      |
|     | 25                                                                                                             | 25.9        | 44.6      | 56.8        | 66.1       | 80.2        | 90.8       | 107.0     |
| 12. | Price to Buy at to Achieve IRR (=PV of Final Year Price), if Net Income Differs by ____, if Terminal PER = 15: |             |           |             |            |             |            |           |
|     | 20.0                                                                                                           | 0.25        | 0.50      | 0.75        | 1.00       | 1.50        | 2.00       | 3.00      |
|     | 25.0%                                                                                                          | \$1.66      | \$3.31    | \$4.97      | \$6.63     | \$9.94      | \$13.25    | \$19.88   |
|     | 30.0                                                                                                           | 1.35        | 2.70      | 4.05        | 5.40       | 8.11        | 10.81      | 16.21     |
|     | 40.0                                                                                                           | 1.11        | 2.22      | 3.33        | 4.44       | 6.66        | 8.88       | 13.32     |
|     | 50.0                                                                                                           | 0.77        | 1.53      | 2.30        | 3.07       | 4.60        | 6.13       | 9.20      |
|     | 60.0                                                                                                           | 0.54        | 1.09      | 1.63        | 2.17       | 3.26        | 4.34       | 6.51      |
|     |                                                                                                                | 0.39        | 0.79      | 1.18        | 1.57       | 2.36        | 3.15       | 4.72      |
| 13. | Net Income in Terminal Year to achieve IRR, at PER:                                                            |             |           |             |            |             |            |           |
|     | 10                                                                                                             | 20.0%       | 25.0%     | 30.0%       | 40.0%      | 50.0%       | 60.0%      |           |
|     | 15                                                                                                             | 1,228,800   | 1,507,041 | 1,833,546   | 2,655,921  | 3,750,000   | 5,178,153  |           |
|     | 20                                                                                                             | 819,200     | 1,004,694 | 1,222,364   | 1,770,614  | 2,500,000   | 3,452,102  |           |
|     | 25                                                                                                             | 614,400     | 753,520   | 916,773     | 1,327,960  | 1,875,000   | 2,589,007  |           |
|     |                                                                                                                | 491,520     | 602,816   | 733,418     | 1,062,368  | 1,500,000   | 2,071,261  |           |
| 14. | Corresponding Sales                                                                                            |             |           |             |            |             |            |           |
|     | 20.0%                                                                                                          |             | 25.0%     | 30.0%       | 40.0%      | 50.0%       | 60.0%      |           |
|     | 10                                                                                                             | 6,291,456   | 7,716,049 | 9,387,754   | 13,598,315 | 19,200,000  | 26,512,144 |           |
|     | 15                                                                                                             | 4,194,304   | 5,144,033 | 6,258,503   | 9,065,544  | 12,800,000  | 17,674,763 |           |
|     | 20                                                                                                             | 3,145,728   | 3,858,025 | 4,693,877   | 6,799,158  | 9,600,000   | 13,256,072 |           |
|     | 25                                                                                                             | 2,516,582   | 3,086,420 | 3,755,102   | 5,439,326  | 7,680,000   | 10,604,858 |           |
| 15. | Rate of Return to <i>This</i> Round, if Purchase at Price =                                                    |             |           |             |            |             |            |           |
|     |                                                                                                                | 0.25        | 0.50      | 0.75        | 1.00       | 1.25        | 1.50       | 1.75      |
|     | 10                                                                                                             | \$0.54      | \$1.09    | \$1.63      | \$2.17     | \$2.71      | \$3.26     | \$3.80    |
|     | 15                                                                                                             | 82.5%       | 58.9%     | 46.5%       | 38.3%      | 32.3%       | 27.5%      | 23.7%     |
|     | 20                                                                                                             | 97.9        | 72.3      | 58.9        | 50.0       | 43.5        | 38.3       | 34.1      |
|     | 25                                                                                                             | 109.6       | 82.5      | 68.3        | 58.9       | 51.9        | 46.5       | 42.1      |
|     |                                                                                                                | 119.2       | 90.8      | 76.0        | 66.1       | 58.9        | 53.2       | 48.5      |

**Exhibit 14** ME2 Financing, Sensitivity Model (Continued)

|     |                                                                          | Year<br>0 | Year<br>1 | Year<br>2  | Year<br>3  | Year<br>4  | Year<br>5  |
|-----|--------------------------------------------------------------------------|-----------|-----------|------------|------------|------------|------------|
| 16. | Terminal Price Necessary to Achieve IRR = ____, if Purchase Price =      |           |           |            |            |            |            |
|     | 20.0%                                                                    | \$0.54    | \$ 1.09   | \$ 1.63    | \$ 2.17    | \$ 2.71    | \$ 3.26    |
|     | 25.0                                                                     | 1.35      | 2.70      | 4.05       | 5.40       | 6.75       | 8.11       |
|     | IRR < 30.0                                                               | 1.66      | 3.31      | 4.97       | 6.63       | 8.28       | 9.94       |
|     | 40.0                                                                     | 2.02      | 4.03      | 6.05       | 8.06       | 10.08      | 12.09      |
|     | 50.0                                                                     | 2.92      | 5.84      | 8.76       | 11.68      | 14.60      | 17.52      |
|     | 60.0                                                                     | 4.12      | 8.25      | 12.37      | 16.49      | 20.61      | 24.74      |
|     |                                                                          | 5.69      | 11.39     | 17.08      | 22.77      | 28.46      | 34.16      |
| 17. | Corresponding Net Income, if Terminal PER = 15                           |           |           |            |            |            |            |
|     | 20.0%                                                                    | \$0.54    | \$ 1.09   | \$ 1.63    | \$ 2.17    | \$ 2.71    | \$ 3.26    |
|     | 25.0                                                                     | 204,800   | 409,600   | 614,400    | 819,200    | 1,024,000  | 1,228,800  |
|     | 30.0                                                                     | 251,173   | 502,347   | 753,520    | 1,004,694  | 1,255,867  | 1,507,041  |
|     | 40.0                                                                     | 305,591   | 611,182   | 916,773    | 1,222,364  | 1,527,955  | 1,833,546  |
|     | 50.0                                                                     | 442,653   | 885,307   | 1,327,960  | 1,770,614  | 2,213,267  | 2,655,921  |
|     | 60.0                                                                     | 625,000   | 1,250,000 | 1,875,000  | 2,500,000  | 3,125,000  | 3,750,000  |
|     |                                                                          | 863,026   | 1,726,051 | 2,589,077  | 3,452,102  | 4,315,128  | 5,178,153  |
| 18. | Corresponding Sales Necessary                                            |           |           |            |            |            |            |
|     |                                                                          | \$0.54    | \$ 1.09   | \$ 1.63    | \$ 2.17    | \$ 2.71    | \$ 3.26    |
|     | 20.0%                                                                    | 1,048,576 | 2,097,152 | 3,145,728  | 4,194,304  | 5,242,880  | 6,291,456  |
|     | 25.0                                                                     | 1,286,008 | 2,572,016 | 3,858,025  | 5,144,033  | 6,430,041  | 7,716,049  |
|     | 30.0                                                                     | 1,564,626 | 3,129,251 | 4,693,877  | 6,258,503  | 7,823,128  | 9,387,754  |
|     | 40.0                                                                     | 2,266,386 | 4,532,772 | 6,799,158  | 9,065,544  | 11,331,930 | 13,598,315 |
|     | 50.0                                                                     | 3,200,000 | 6,400,000 | 9,600,000  | 12,800,000 | 16,000,000 | 19,200,000 |
|     | 60.0                                                                     | 4,418,691 | 8,837,381 | 13,256,072 | 17,674,763 | 22,093,453 | 26,512,144 |
| 19. | PERs Implicit in Terminal Valuations Based on Cap. Rate and Growth Rates |           |           |            |            |            |            |
|     | Growth Rates —                                                           | 2.5%      | 5.0%      | 7.5%       | 10.0%      | 12.5%      | 15.0%      |
|     | 5.0%                                                                     | 37        | NA        | NA         | NA         | NA         | NA         |
|     | 10.0                                                                     | 12        | 17        | 31         | NA         | NA         | NA         |
|     | Cap. Rates < 15.0%                                                       | 7         | 9         | 10         | 14         | 26         | NA         |
|     | 20.0                                                                     | 5         | 6         | 6          | 7          | 9          | 12         |
|     | 25.0                                                                     | 4         | 4         | 4          | 5          | 5          | 6          |
|     | 30.0                                                                     | 3         | 3         | 3          | 4          | 4          | 4          |
| 20. | Multipliers Corresponding to Terminal PER =                              |           |           |            |            |            |            |
|     | 5                                                                        | 10        | 15        | 20         | 25         | 30         |            |
|     | Revenues                                                                 | 1.0       | 2.0       | 2.9        | 3.9        | 4.9        | 5.9        |
|     | Year 0 Price                                                             | 2.5       | 3.1       | 4.7        | 6.3        | 7.8        | 9.4        |
|     | Book Value                                                               | 1.6       | 3.1       | 4.7        | 6.2        | 7.8        | 9.3        |
|     | Total Assets                                                             | 1.4       | 2.8       | 4.3        | 5.7        | 7.1        | 8.5        |

### Appendix 1 Derivation of Formulae for Shares Issued

Given a situation in which an investor has agreed to buy a certain *percent ownership* in a company with a certain number of shares (*old shares*) outstanding, how many *new shares* must be issued?

1. By definition:  $\text{current shares} = \text{old shares} + \text{new shares}$
2. By definition:  $\% \text{ ownership acquired} = \frac{\text{new shares}}{\text{current shares}}$
3. From (1):  $\% \text{ ownership} = \frac{\text{new shares}}{\text{old} + \text{new shares}}$
4.  $(\% \text{ own} \times \text{old}) + (\% \text{ own} \times \text{new}) = \text{new shares}$
5. Solving:  $\text{new shares} = \frac{\% \text{ ownership}}{1 - \% \text{ ownership}} \times \text{old shares}$

The above formula can also be expressed in terms of total current shares. Start by adding *old shares* to each side:

6.  $\text{old} + \text{new shares} = \left( \frac{\% \text{ ownership}}{1 - \% \text{ ownership}} \times \text{old shares} \right) + (\text{old shares})$
7. From (1)  $\text{current shares} = \left( \frac{\% \text{ ownership}}{1 - \% \text{ ownership}} + 1 \right) \times (\text{old shares})$
8.  $= \left( \frac{\% \text{ own}}{1 - \% \text{ own}} + \frac{1 - \% \text{ own}}{1 - \% \text{ own}} \right) \times (\text{old shares})$
9.  $\text{current shares} = \frac{1}{1 - \% \text{ ownership}} \times (\text{old shares})$

**Appendix 2** Derivation of Formula for Retention and Dilution Working with Retention Percent

Retention percent is the key to converting a VC's required *final ownership* (the percent of terminal value (TV) that he must own to realize his required return on investment) into the required *current ownership* (the percent of the company he must buy now). It is defined as:

$$1. \text{ retention \%} = 1 - \text{total \% of TV to be sold to other owners}$$

where the *other owners* includes future investors and new managers. We shall see that retention % can also be thought of as *the percent of total TV available to the current investors*. Let us define:

$$2. \text{ final shares} = \text{current shares} + \text{additional future shares}$$

where *current shares* and *final shares* refer, respectively, to the total number of shares outstanding at the end of the current financing and to the total number of shares outstanding in the terminal year.

$$3. \text{ From (1), } \text{retention \%} = 1 - \frac{\text{additional future shares}}{\text{final shares}}$$

$$4. \quad = \frac{(\text{final shares}) - (\text{additional future shares})}{\text{final shares}}$$

$$5. \text{ From (2) } \text{retention \%} = \frac{\text{current shares}}{\text{final shares}}$$

Let *new shares* equal the number of shares that the investor will purchase in the current financing.

$$6. \quad \text{retention \%} = \frac{\text{current shares}}{\text{final shares}} \times \frac{\text{new shares}}{\text{new shares}}$$

$$7. \quad = \frac{(\text{new shares}) / (\text{final shares})}{(\text{new shares}) / (\text{current shares})}$$

$$8. \quad \text{retention \%} = \frac{\text{final ownership \%}}{\text{current ownership \%}}$$

$$9. \quad \text{current ownership \%} = \frac{\text{final ownership \%}}{\text{retention \%}}$$

Recall the basic venture capital valuation formula:

$$10. \text{ required final ownership \%} = \frac{\text{final value (investment)}}{\text{terminal value}}$$

## Appendix 2 Derivation of Formula for Retention and Dilution Working with Retention Percent (Continued)

Multiplying each side by  $1/(\text{retention } \%)$ , and applying formula (9) to the left side:

$$11. \text{ required final ownership } \% = \frac{\text{final value (investment)}}{(\text{terminal value}) \times (\text{retention } \%)}$$

Note that formula (10) uses the total TV, and calculates a final ownership percent which must then be adjusted to a current ownership share. Formula (11), which calculates the current ownership percent, refers only to that percent of the terminal value which will not be sold to future investors, and which is therefore available to the current investors. Although it is convenient to calculate the current ownership in one step, it is necessary to project each investor's final ownership percentage before separately converting to current ownership shares, in order to determine the final percent available for management or for unexpected changes in financing.

### *Retention versus Dilution*

Many venture capitalists work with *dilution percent*, which is the *ratio of future additional shares to current shares*. Like retention percent, it can also be used to convert future ownership to current:

$$12. \quad \text{dilution } \% = \frac{\text{future additional shares}}{\text{current shares}}$$

$$13. \quad \text{From (2) } 1 + \text{dilution } \% = \frac{\text{current} + \text{additional shares}}{\text{current shares}}$$

$$14. \quad = \frac{\text{final shares}}{\text{current shares}} \times \frac{1 / (\text{new shares})}{1 / (\text{new shares})}$$

$$15. \quad = \frac{(\text{new shares}) / (\text{current shares})}{(\text{new shares}) / (\text{final shares})}$$

$$16. \quad \text{dilution } \% = \frac{\text{current ownership}}{\text{final ownership}} - 1$$

$$17. \quad \text{current ownership } \% = (\text{final ownership}) \times (1 + \text{dilution } \%)$$

By formula (9), it is clear that:

$$18. \quad 1 + (\text{dilution } \%) = 1 / (\text{retention } \%)$$

$$19. \text{ So, } \quad \text{dilution } \% = \frac{1}{\text{retention } \%} - 1$$

$$20. \text{ And, } \quad \text{retention } \% = \frac{1}{1 + \text{dilution } \%}$$

## Appendix 2 Derivation of Formula for Retention and Dilution Working with Retention Percent (Continued)

As the following table of equivalent values indicates, dilution and retention are inversely related:

|             |      |     |     |     |     |     |
|-------------|------|-----|-----|-----|-----|-----|
| Retention % | 100% | 80% | 67% | 50% | 33% | 25% |
| Dilution %  | 0    | 25  | 50  | 100 | 200 | 300 |

There is a circularity problem inherent in using dilution rather than retention. Calculating dilution using formula (12) requires knowledge of current (post-financing) shares and of additional future shares. In order to calculate current shares, we must know the investor's required current ownership stake. On the other hand, in order to derive required current ownership share from required final ownership percentage, we need an estimate of dilution or retention. Thus, current ownership is needed to calculate dilution, but dilution is needed to calculate current ownership.

Dilution percent, defined with current ownership as its denominator, looks backward. The valuation method presented in this paper uses retention percent, which is defined looking forward, using terminal ownership shares.

### Appendix 3 “The First Chicago Method”

Although the method presented above has proved successful for many venture capitalists, it should be clear that the use of the high, risk-adjusted, discount rate embodies many assumptions. In particular, the use of this high discount rate has evolved to compensate for the fact that subsequent performance of individual companies does not always match the projections contained in the original plans.

As has been demonstrated above (see “The Venture Capitalist’s Discount Rate: cash flow adjustment”), for a certain pattern of cash returns, a 50% rate for first-stage ventures can compensate for that half of ventures that only provide a return of capital plus moderate dividends (the sideways scenario), and for that quarter of ventures that lose most of the capital invested.

With comparable ventures, this system will work smoothly, but when comparing ventures with very different cash flow characteristics, a bias is created. For example, consider a potential investment that offers a less dramatic success scenario, but which is capable of marginally acceptable returns in the sideways scenario or a near-complete return of capital under liquidation in the failure scenario. This venture will be discriminated against under the traditional venture capital method presented in this note, even if it offers an equivalent expected cash flow overall. The traditional method implicitly assumes that every venture offers the same relative cash flows under the unfavorable scenarios.

One alternative valuation method, developed at First Chicago Corp.’s venture capital group, employs a lower discount rate, but applies it to an *expected* cash flow. That expected cash flow is the average of three possible scenarios, with each scenario weighted according to its perceived probability. We examine the MicroElectronics Enterprises example again, from the point of view of the first round venture capitalist who invests \$1,500,000. The investment is in the form of 13% redeemable preferred stock.

**The Success Scenario (IPO):** This is often the only scenario that is examined under the traditional method. Under this scenario, the VC will collect a 13% dividend for four years, and then capture his share of an IPO in the fifth year. We don’t yet know what final ownership percent will be required, so the fifth year’s cash flow must be stated as a formula:

$$(\text{accrued dividends}) + ((\text{final ownership \%}) \times (\text{terminal value}))$$

Both accrued dividends as of IPO and terminal value can be estimated now, but the final ownership must be left as a variable. We will solve for that ownership percent below.

**The Sideways Scenario:** In this case, the company survives with enough profitability to pay dividends for three years, after which a four-year straight-line redemption begins, yielding the VC a 13% rate of return. As this method makes apparent, given the high probability that the venture will at least live out this scenario, it is important for the VC to structure his investment so as to return some value in the absence of an IPO.

**The Failure Scenario (liquidation):** The cash flows under this scenario will vary greatly according to the expected use of the funding. For a service venture that applies its entire capitalization toward salaries, there will be no opportunity to recover capital (other than remaining cash) under liquidation. For a distributorship or retailing venture that applies its capital to working capital, much of the committed capital should be recoverable. For the case of ME2, we assume that in its final year of life the company pays half of the expected dividends and that 20% of the cash invested is recovered upon liquidation.

### Appendix 3 “The First Chicago Method” (Continued)

Once the scenarios are laid out, the probability of each must be determined. This task, of course, cannot be performed with accuracy or with total confidence. However, the VC is also assigning probabilities of success and failure under the traditional method, when he adopts a high, risk-adjusted discount rate. Under this method, those expected probabilities are made explicit.

For the ME2 example, we assume that the probabilities are 25%/50%/25% as has proved typical of venture capital investments.

|               |         |                     |          |
|---------------|---------|---------------------|----------|
| Year of IPO   | 5       | Terminal Net Income | \$ 2,500 |
| Discount Rate | 28.0%   | Terminal PER        | 15       |
| Investment    | \$1,500 | Terminal Value      | \$37,500 |
| Dividend Rate | 13.0%   |                     |          |

| Scenario | p    | 1     | 2     | 3     | 4     | 5          | 6     | 7     |
|----------|------|-------|-------|-------|-------|------------|-------|-------|
| Up       | 0.25 | \$195 | \$195 | \$195 | \$195 | \$150+X    | \$0   | \$0   |
| Sideways | 0.50 | 195   | 195   | 195   | 504   | 504        | 504   | 504   |
| Down     | 0.25 | 195   | 400   | 0     | 0     | 0          | 0     | 0     |
| WGT AVG  |      | \$195 | \$246 | \$146 | \$301 | \$290+.25X | \$252 | \$252 |

Where  $X = (\text{final ownership}\%) \times (\text{terminal value})$

The weighted average cash flow above represents the *expected* cash flow. Since the cash flow has been explicitly adjusted above, it would be redundant to penalize it again by employing a traditional high discount rate (such as the 50% used in the ME2 example). The VC will discount the expected cash flow at the expected portfolio average discount rate of, say, 28%.

The basic venture capital formula, presented at the beginning of this note, was derived from the insight that one could focus on future values when calculating required current ownership shares. In the traditional method, the future value of the VC's investment had to equal to his portion of the terminal value. In the First Chicago method, the VC must account for value received from harvest and also from the other components of the expected cash flow above:

$$FV(\text{expected cash flow}) = FV(\text{investment})$$

The expected cash flow can be broken down into the expected cash from IPO (.25 x X) and the rest of the cash flow (the “other cash flow”).

$$(.25 \times (\text{final ownership}) \times TV) + FV(\text{non-IPO cash flow}) = FV(\text{investment})$$

Note that the .25 above is the probability of the success scenario. Solving for final ownership, we wind up with a modification of the basic venture capital formula:

$$\text{Required final ownership} = \frac{FV(\text{investment}) - FV(\text{non-IPO cash flow})}{p(\text{success}) \times (\text{forecast terminal value})}$$

The formula above differs from the original basic venture capital formula in two ways: First, the basic formula assumes that there are no cash flows between the investment and the harvest in year 5.

### Appendix 3 “The First Chicago Method” (Continued)

Above, the future value of the intermediate cash flows is subtracted from the future value of the investment, because it is the difference between them that must be made up for out of the terminal value. Second, the basic formula does not distinguish between the *forecast* terminal value and the *expected* TV. The traditional method uses the forecast TV, which is adjusted through the use of a high discount rate. Above, the expected value of the TV is employed. Since the success (IPO) scenario has a 25% probability, the expected value of the TV is equal to (forecast terminal value) x .25.

The required final ownership in ME2 can now be calculated. Note that the “future” values are as of the IPO in year 5. Since the expected (weighted average) cash flow continues through year 7, the easiest way to calculate the year-5 value is to first compute the net present (year-0) value of the cash flow, and then convert that NPV to a year-5 value by multiplying by 1.28<sup>5</sup>.

$$\begin{aligned}\text{required final ownership} &= \frac{\$1.5\text{mm} \times 1.285 - \text{NPV}(\text{non-IPO cash}) \times 1.285}{\$2.5\text{mm} \times 15 \times .25} \\ &= \frac{\$5.2\text{mm} - \$2.3\text{mm}}{\$37.5\text{mm} \times .25} \\ &= 30.4\%\end{aligned}$$

which equals, not coincidentally, the required final ownership that was derived using the traditional method.

As with the method presented in the note, the investor must now repeat the above process for each of the following rounds that are expected. Under the traditional method that required forecasting the discount rates that future investors would require, those rates generally are lower in each ensuing round. With this method, however, the discount rate used will remain essentially the same, but the probability of success will improve in later rounds. After each investor’s required final ownership is projected, the retention rate, the required current ownership, the number of shares to issue, and the share price can all be calculated using the formulas presented in this note.

In general, applying a scenario-linked forecasting and valuation method will be very useful when the payoffs under each scenario differ widely from the norm. Thus, if the company has the ability to generate substantial value in liquidation, this fact must be noted and taken into account when valuing the deal. Such an investment will be more valuable than one with the same characteristics but a low liquidation value, a fact that application of the traditional method would not reveal.

## Appendix 4 The Term Structure of Discount Rates

One of the most crucial determinants of pricing is the set of discount rates that the VC determines for himself and for future rounds. As the VC projects rates for the future rounds, he must put himself in the position of the future investor (indeed, he is likely to be the future investor), and consider the rate that he would demand if he were in that year-two or year-four position. By examining the relationship between the rates for the various rounds, the VC can confront himself with some assumptions he is making about the behavior of the venture's valuation of the early years.

In the example above, the VC demands a 50% rate in year 0, and projects that a 40% rate would be acceptable in year 2. Selecting a 50% rate means that the investment should grow at a 50% average rate over five years. However, during years three through five, both the year-0 and the year-2 investments will be exposed to the same risk. Thus, the difference in rates reflects the higher risk of the first two years. There is a strong chance, as of the first round, that the second round either will never happen (due to the failure of the company), will happen under crisis conditions (allowing the second-round VCs to negotiate any price they desire), or will be subject to a less sanguine forecast (in which case lowered terminal value expectations will motivate a lower price).

If the VC would accept a 40% return over the final three years, what is the rate of growth in value that he is demanding for the first two years? This VC is expecting the shares he buys at \$2.91 to grow to \$8.06 over two years, implying a growth rate of 66.4% over that period.

A way to compute the same result directly from the discount rates is to calculate the so-called geometric average of the interest rates. If the initial two years' rate is  $X\%$ :

$$\begin{aligned}(1 + 50\%)^5 &= (1 + X\%)^2 \times (1 + 40\%)^3 \\ X\% &= \left( \frac{(1 + 50\%)^5}{(1 + 40\%)^3} \right)^{1/2} - 1 \\ &= 66.4\%\end{aligned}$$

The various short-term discount rates implicit in the ME2 projections are summarized in **Exhibit 3**, part 7. We can therefore examine the *term structure* of the projected discount rates. For example, that table shows that the year-zero VC implicitly expects his investment to grow at 66.4% over the next two years, and at 57.0% over the next four years.

On the one hand, the rates computed above are not meaningful rates of return. While the first-round VC expects that the second-round VC will buy stock at \$8.06, he also knows that he will not have the option of selling out at that price. A potential buyer would be concerned about the lack of confidence demonstrated by a VC who is looking to be bailed out of his investment. Furthermore, a VC buying new stock from a company knows that his investment will be adding value to that company. If he were to buy another VC's stock, the company's net worth would remain unchanged, and its capital position would remain as desperate as it might have been before the stock purchase.

On the other hand, examining the term structure of the projected discount rates can be a valuable test of the plausibility of the VC's expectations. Consider a VC who invests using a discount rate of 50% over five years, expecting that the next round will occur one year later with a discount rate of only 35%. Using the geometric average formula from above, the implicit rate over the first year is:

$$\left( \frac{(1 + 50\%)^5}{(1 + 35\%)^4} \right)^{1/1} - 1 = 128.6\%$$

This VC should have a strong argument for why the next round's VCs would assign over twice the value to his shares. Such a jump in valuation is not necessarily unreasonable; perhaps, for example, the company is attempting to develop a proprietary technology and the first year will reveal whether or not it will have the potential to dominate the industry. If the technology development is successful, the 35% rate will be reasonable.<sup>14</sup>

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<sup>14</sup> Some venture capitalists examine the step-up from round to round. The step-up is the ratio of one round's share price to the previous round's price. For ME2, the step-up in round 2 is 2.8, in round 3: 2.2, and for the IPO: 1.3. Although step-up can serve as a quick check on relative pricing and as a useful calculation tool, it remains a backward-looking measurement, in that it compares a price to previous rounds, rather than to future values. Of course, step-up is further flawed as an analytical tool in that it ignores the time value of money: A step-up of 2 across a period of one year yields an implicit rate of return of 100% to the first round investor for that year. Across three years, that step-up implies only a 25% rate.